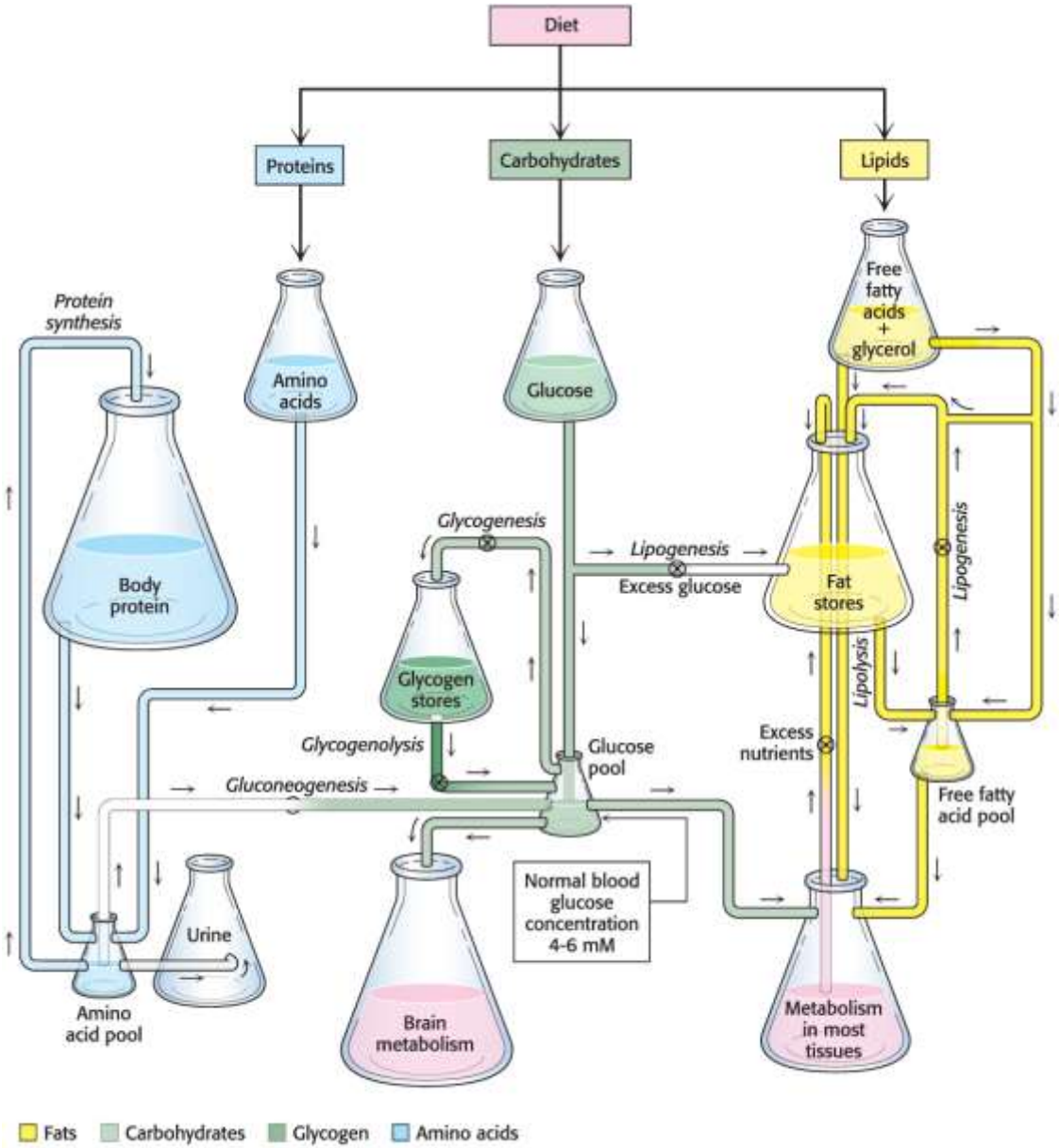
21BTC101T - Biochemistry

Unit – II – Introduction to Metabolism, Bioenergetics and Photosynthesis

- **Introduction to Metabolism**
- **Bioenergetics, High energy compounds, Biological oxidation**
- **Electron transport chain, Oxidative phospholyration, Chemiosmotic theory**
- **Shuttle pathway – Glycerol phosphate Shuttle, Malate aspartate Shuttle**
- **Photosynthesis and Light reaction**

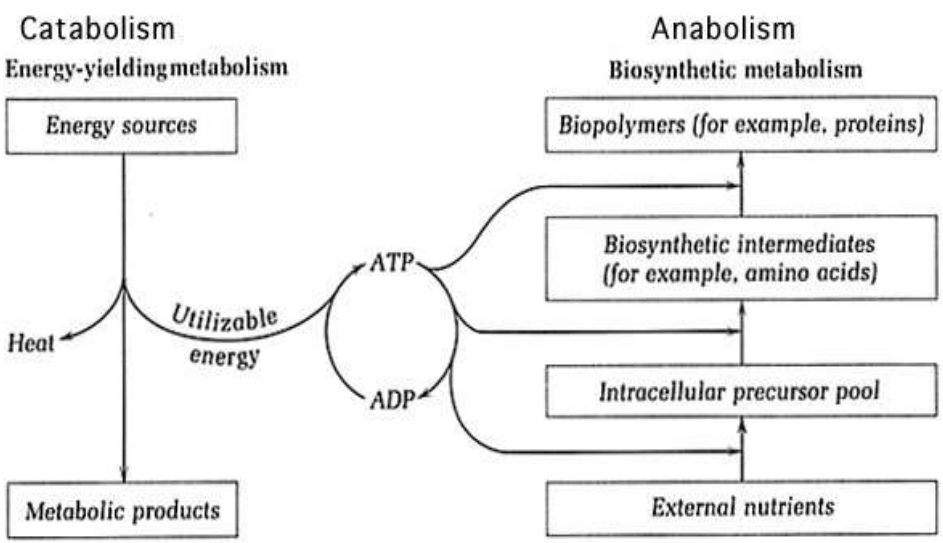
Diagram of Homeostasis

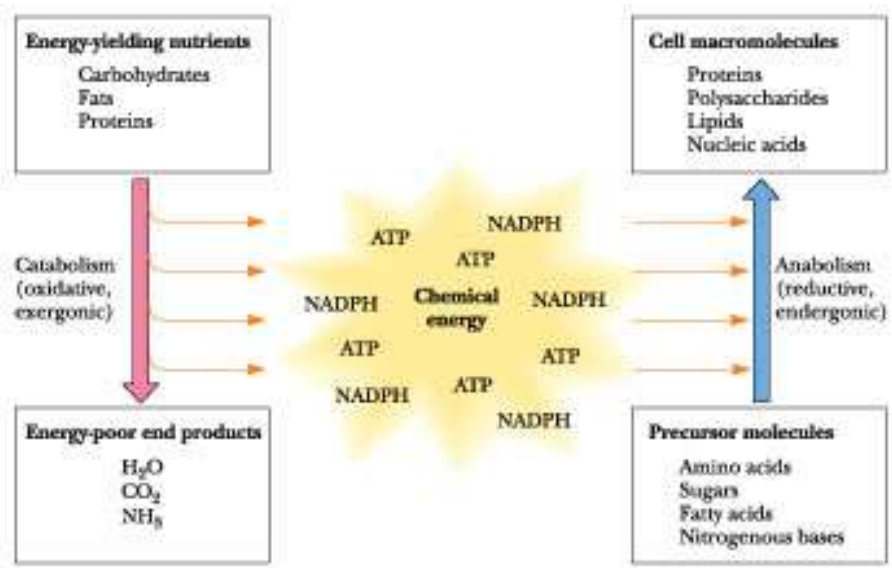


Tymoczko et al., *Biochemistry: A Short Course*, 4e, © 2019 W. H. Freeman and Company

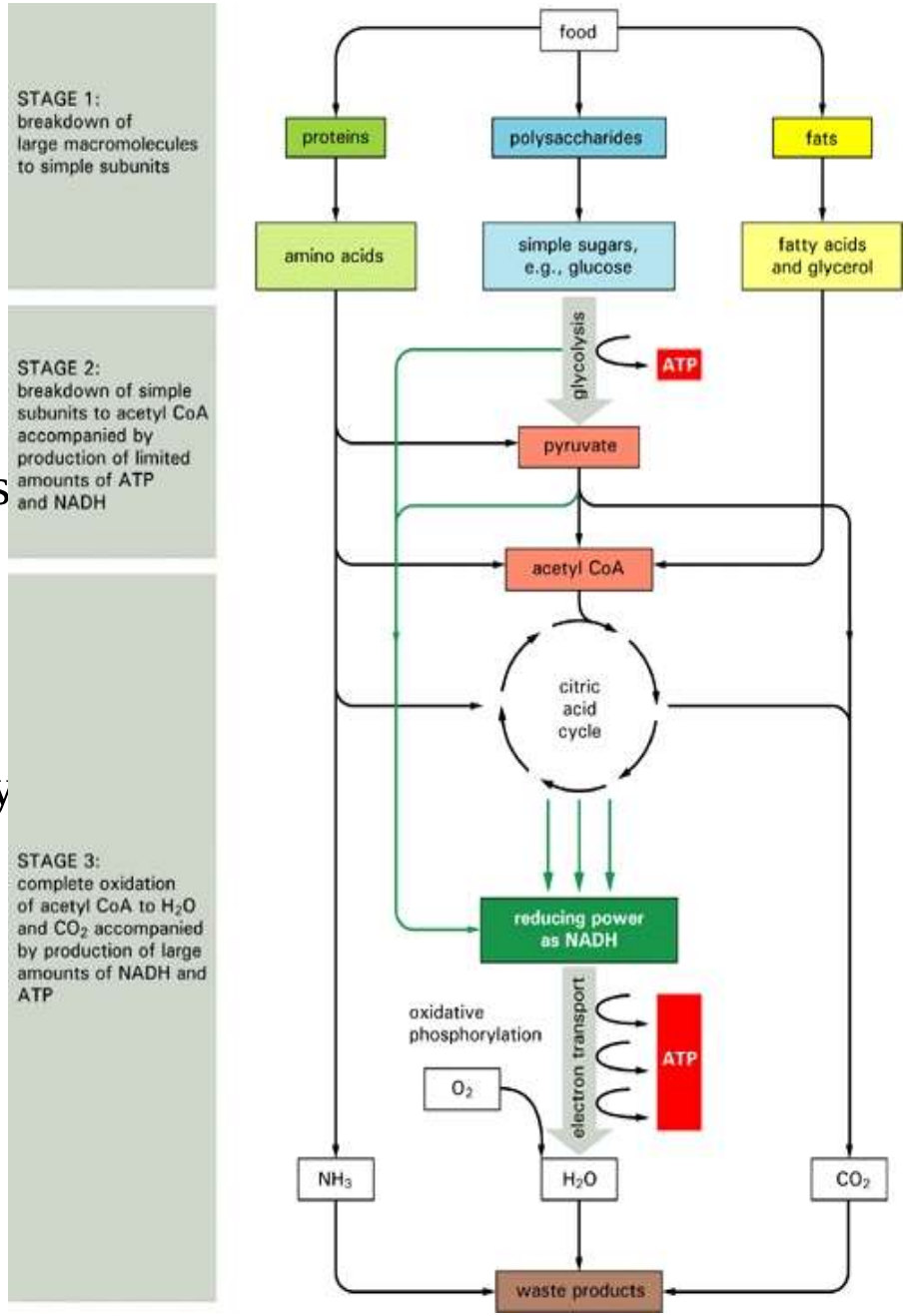
Introduction to Metabolism

- **Complex substances are broken down for energy, required metabolites, structural components, etc.**
- **Cells must synthesize new complex substances.**
- **Thousands of such reactions are occurring simultaneously in a single cell.**
- **These reactions occur with a minimum of side products, energy loss or undesired interferences and at reasonable temperatures, pH, and pressure.**
- **All of these reactions must be controlled or regulated for optimum efficiency.**





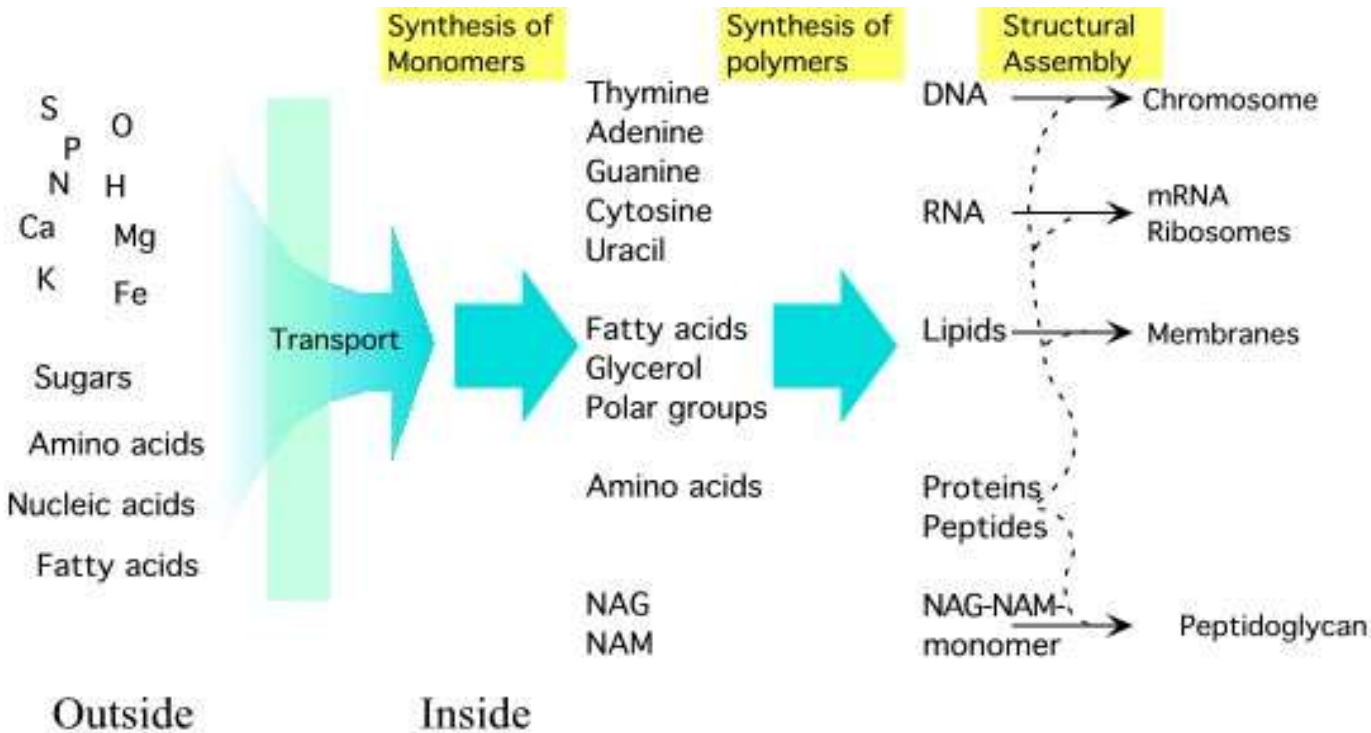
- **Metabolism** = all the chemical reactions that occur in an organism
- **Catabolism** = the breakdown of complex substances.
- Cellular metabolism
  - Cells *break down* excess carbohydrates first, then lipids, finally amino acids if energy needs are not met by carbohydrates and fat
  - Nutrients not used for energy are used to *build up* structure, are stored, or they are excreted
  - 40% of the energy released in catabolism is captured in ATP, the rest is released as heat

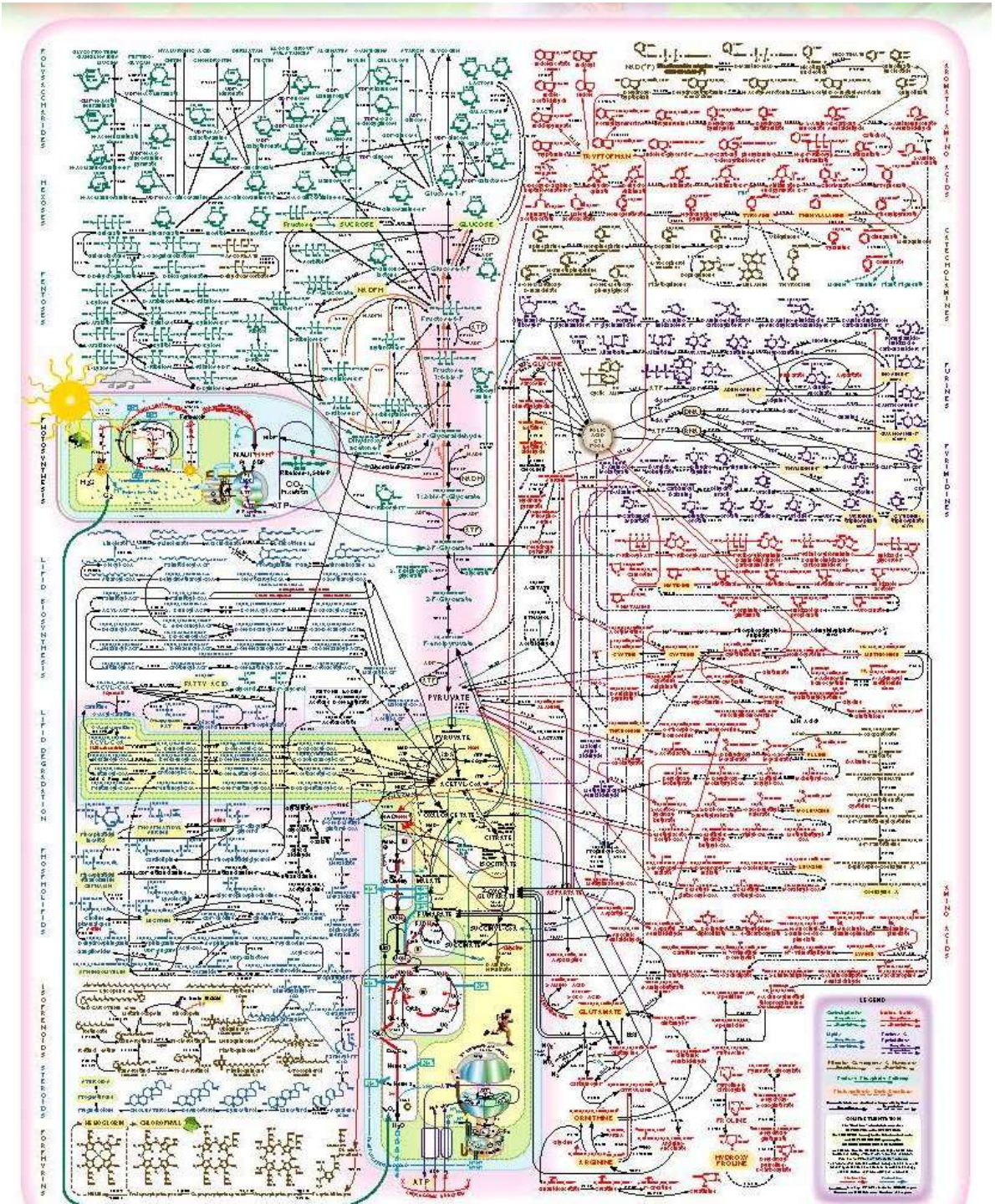


- **Anabolism** = the synthesis of complex substances from simpler ones.
  - **Performance of structural maintenance and repairs**
  - **Support of growth**
  - **Production of secretions**
  - **Building of nutrient reserves**
  - **Starting materials are *pyruvate, acetyl CoA, and the intermediates of citric acid cycle***
  - **Dependent on the *supply of energy* (ATP/GTP) and *reducing equivalents* (NADPH+, H+)**

NADPH:

- *reduced form* of nicotinamide adenine dinucleotide phosphate.
- a coenzyme used in anabolic reactions, such as lipid and nucleic acid synthesis

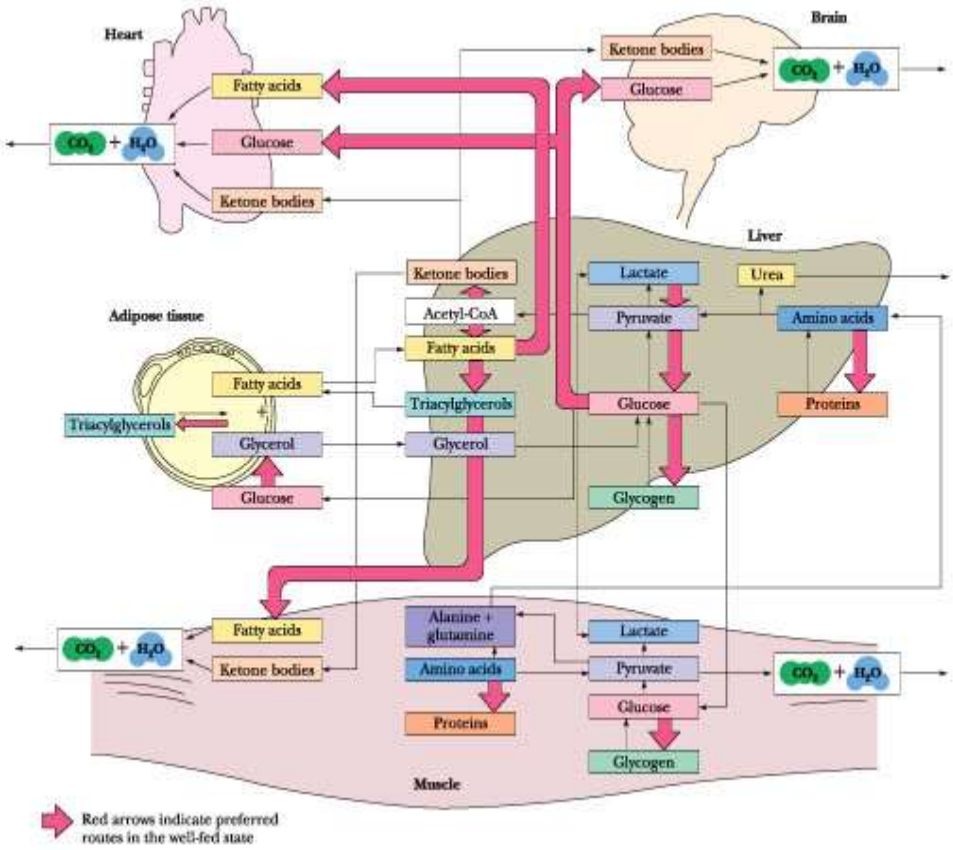




A metabolic map, indicating the reactions of intermediary metabolism and the enzymes that catalyze them. Over 500 different chemical intermediates, or metabolites, and a greater number of enzymes are represented here.

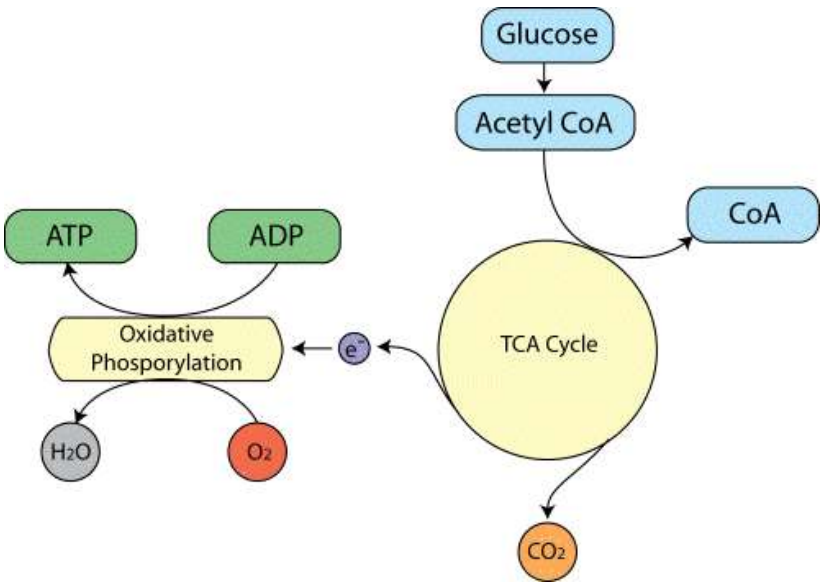
Metabolic relationships among the major human organs: brain, muscle, heart, adipose tissue, and liver

Organ	Energy Reservoir	Preferred Substrate	Energy Sources Exported
Brain	None	Glucose (ketone bodies during starvation)	None
Skeletal muscle (resting)	Glycogen	Fatty acids	None
Skeletal muscle (prolonged exercise)	None	Glucose	Lactate
Heart muscle	Glycogen	Fatty acids	None
Adipose tissue	Triacylglycerol	Fatty acids	Fatty acids, glycerol
Liver	Glycogen, triacylglycerol	Amino acids, glucose, fatty acids	Fatty acids,glucose, ketone bodies

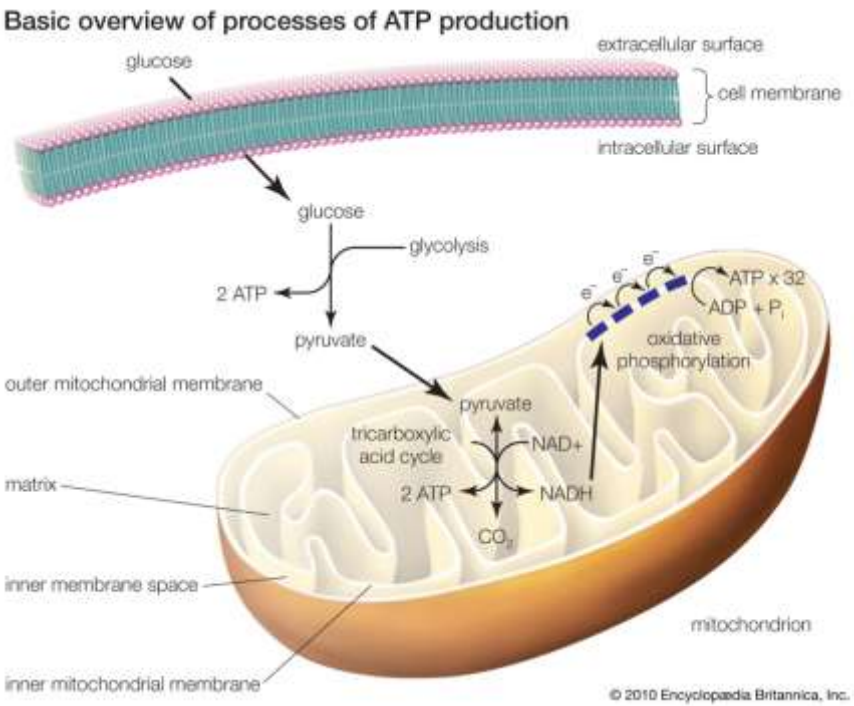


Introduction to Bioenergetics

- Will study in glycolysis and the TCA cycle convert some of the energy available from stored and dietary sugars directly to ATP.
- However, most of the metabolic energy that is obtainable from substrates entering glycolysis and the TCA cycle is funneled via oxidation-reduction reactions into NADH and reduced flavoproteins, the latter symbolized by [$FADH_2$].
- Now we will observe how cells convert the stored metabolic energy of NADH and [$FADH_2$] into ATP.
- Whereas ATP made in glycolysis and the TCA cycle is the result of substrate-level phosphorylation, NADH-dependent ATP synthesis is the result of **oxidative phosphorylation.**



- Electrons stored in the form of the reduced coenzymes, NADH or [$FADH_2$], are passed through an elaborate and highly organized chain of proteins and coenzymes, the so-called **electron transport chain,** finally reaching O_2 (molecular oxygen), the terminal electron acceptor.
- Each component of the chain can exist in (at least) two oxidation states, and each component is successively reduced and reoxidized as electrons move through the chain from NADH (or [$FADH_2$]) to O_2.
- In the course of electron transport, a proton gradient is established across the inner mitochondrial membrane. It is the energy of this proton gradient that drives ATP synthesis.



Bioenergetics

- **Bioenergetics or biochemical thermodynamics** deals with the transformation and use of energy by living cells.
  - The biochemical reations are associated with the liberation of energy in the form of heat.
  - reactions are broadly classified into **exergonic (energy releasing)** and **endergonic (energy consuming)**
- Bioenergetics is concerned with the initial and final state of energy component of the reactants and not the mechanism of chemical reactions
- Energy actually available to do work (utilizable) is known as **free energy**
  - **Changes in free energy (ΔG)** is useful in predicting the feasibility of chemical reactions
  - **Enthalpy (ΔH)** is a measure of the change in heat content of the reactants compared to products
  - **Entropy (ΔS)** is a change in the randomness or disorder of reactants and products
- The relationship between the above three is indicated by the equation,
  - $\Delta G = \Delta H - T\,\Delta S$ where T is the absolute temperature in Kelvin ([K] = [°C] + 273.15)
  - **standard free energy** ($\Delta G°$) is the free energy change when the reactants or products are at a concentration of 1 mol/l at pH 7.0

High energy compounds

- Certain compounds in the biological system which, on hydrolysis, yield energy of at least 7 Cal/mol at pH 7.0 are referred to as **high energy compounds**
  - any other compounds liberating energy below this amount (lower than ATP hydrolysis to ADP + Pi) are referred to as **low energy compounds**
- All the high energy compounds when hydrolyzed liberate more energy than that of ATP.
- Most of high energy compounds contain phosphate group (exception acetyl CoA) hence they are also called high energy phosphates.

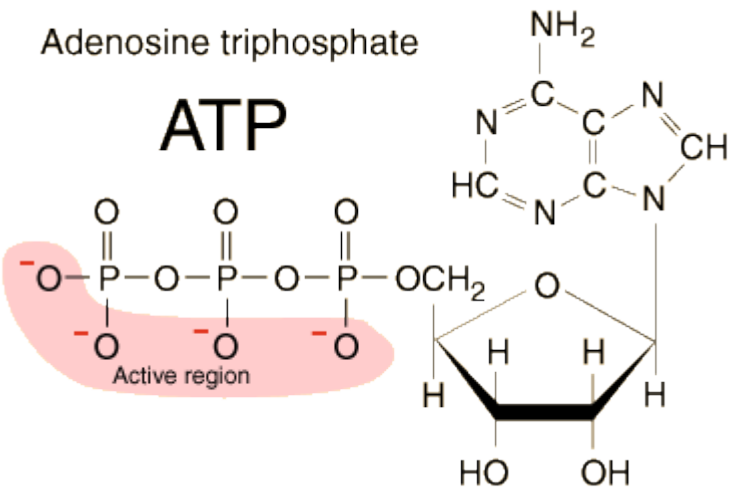
Compounds	ΔG^o (Cal/mol)
High – Energy Phosphates	
Phosphoenol pyruvate	- 14.8
Carbamoyl phosphate	- 12.3
Cyclic AMP	- 12.0
1,3 – Bisphosphoglycerate	- 11.8
Phosphocreatine	- 10.3
Acetyl phosphate	- 10.3
S – Adenosylmethionine	- 10.0
Pyrophosphate	- 8.0
Acetyl CoA	- 7.7
ATP→ADP + Pi	- 7.3
Low energy compounds	
ADP→AMP + Pi	- 6.6
Glucose 1-Phosphate	- 5.0
Fructose 6-Phosphate	- 3.8
Glucose 6-Phosphate	- 3.3
Glycerol 3-Phosphate	- 2.2

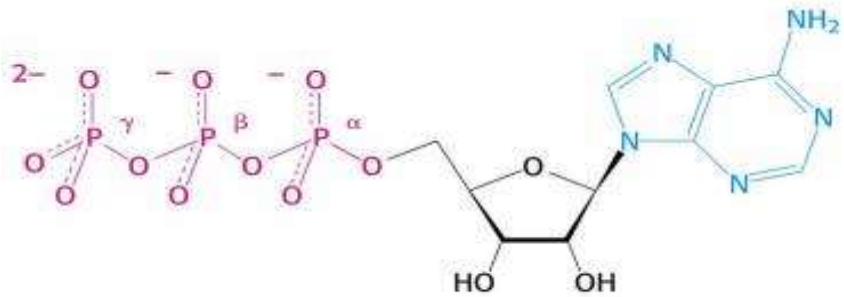
- **High – energy bonds:** The high energy compounds possess ***Acid anhydride bonds*** (mostly phosphoanhydride bonds) which are formed by the condensation of two acidic groups or related compounds. These bonds are referred to as high energy bonds, since the free energy is liberated when these bonds are hydrolyzed. Lipmann suggested use of the ***symbol ~ to represent high energy bond***. For instance, ATP is written as AMP ~ P ~ P.

Class	Bond	Example (s)
Pyrophosphates	– C – P – P	ATP, pyrophosphate
Acyl phosphates	O ‖ – C – O ~ P	1,3-Bisphosphoglycerate, Carbamoyl phosphate, Acetyl phosphate.
Enol phosphates	– CH ‖ – C – O ~ P	Phosphoenol pyruvate
Thiol esters (thioesters)	C ‖ – C – O ~ S –	Acetyl CoA, Acyl CoA
Guanido phosphates (phosphagens)	| – N~ P	Phosphocreatine, Phosphoarginine

ATP – the most important high energy compound

- Adenosine triphosphate (ATP) is a unique and the most important high energy molecule in the living cells.
- The ATP molecule is a purine (adenine) nucleotide in which the adenine is attached in a glycosidic linkage to D – ribose.
- Three phosphoryl groups esterified to the 5 position of the ribose moiety in phosphoanhydride bonds.
- **The two terminal phosphoryl groups (i.e., β and γ) are involved in the phosphoric acid anhydride bonding and are designated as energy rich or high energy bonds**

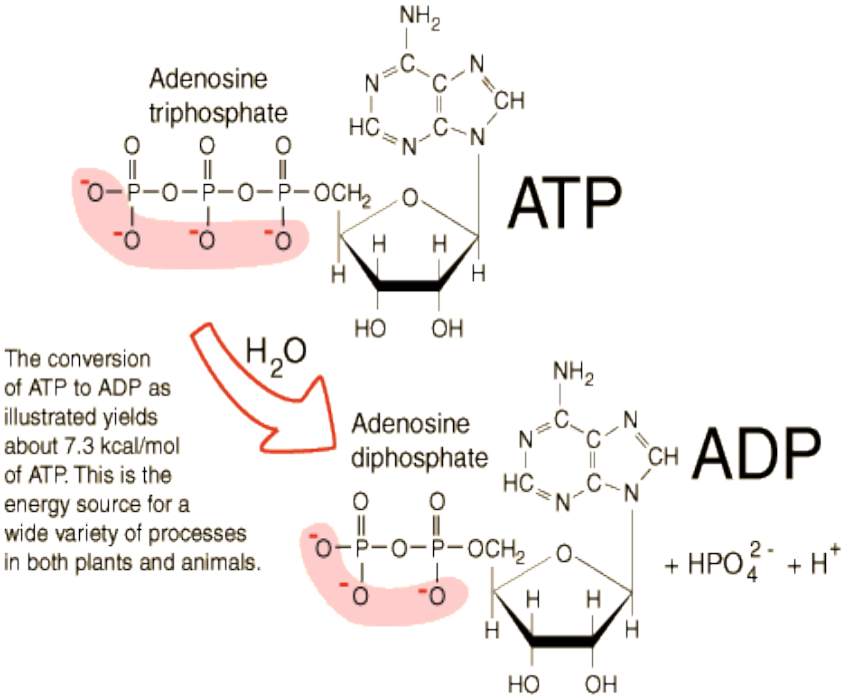




- ATP serves as the ***energy currency of the cell*** as is evident from the ATP – ADP cycle.
- As a result of its position midway down the list of standard free energies of hydrolysis, ATP is able to act as donor of high energy phosphate to those compounds below it in the table.
- Likewise, provided the necessary enzymatic machinery is available, ADP can accept high energy phosphate to form ATP from those compounds above ATP in the table. The hydrolysis of ATP is associated with the large amount of energy.

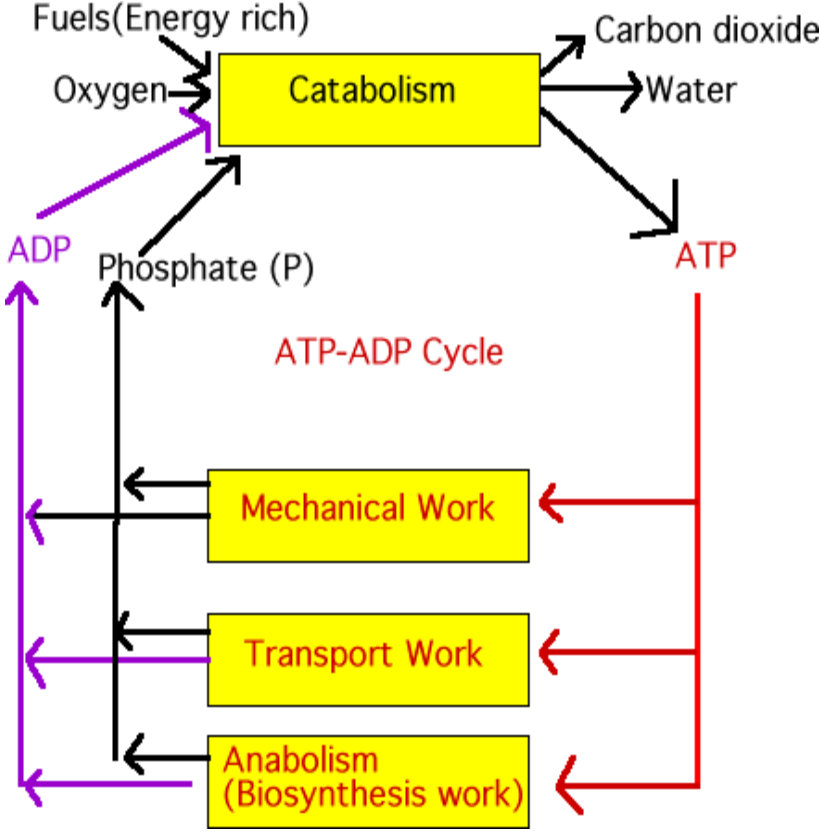
$$ATP + H_2O \rightarrow ADP + Pi$$

$$\Delta G^{o\prime} = -7.3 \text{ cal. mol}^{-1}$$



The conversion of ATP to ADP as illustrated yields about 7.3 kcal/mol of ATP. This is the energy source for a wide variety of processes in both plants and animals.

- The energy liberated is utilized for various processes like muscle contraction, active transport etc.
- In effect, an ATP – ADP cycle connects these processes that generate ~P to those processes that utilize ~P.
- Thus, ATP is continuously consumed and regenerated. This occurs at a very rapid rate, since the total ATP/ADP pool is extremely small and sufficient to maintain an active tissue only for a few seconds.
- ATP acts as an ***energy link between Catabolism*** (degradation of molecules) and ***Anabolism*** (synthesis) in the biological system.



- ATP can be synthesized in 2 ways:

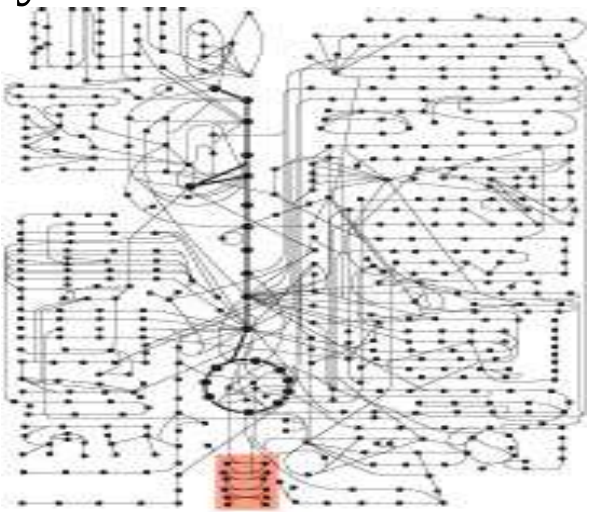
  1) ***Oxidative phosphorylation***. This is the greatest quantitative source of ~P in aerobic organisms. The free energy to drive this process comes from respiratory chain oxidation within mitochondria.

  2) ***Substrate level phosphorylation***. ATP formed during substrate oxidation. Phosphoenolpyruvate and 1,3 bisphosphoglycerate (glycolysis) and succinyl CoA (TCA) can transfer high-energy phosphate to ATP.

- Vertebrates: Phosphocreatine (creatine phosphate) stored in muscle and brain is energy-rich compound.
- Invertebrates: Phosphoarginine (arginine phosphate) is stored.

Electron Transport Chain

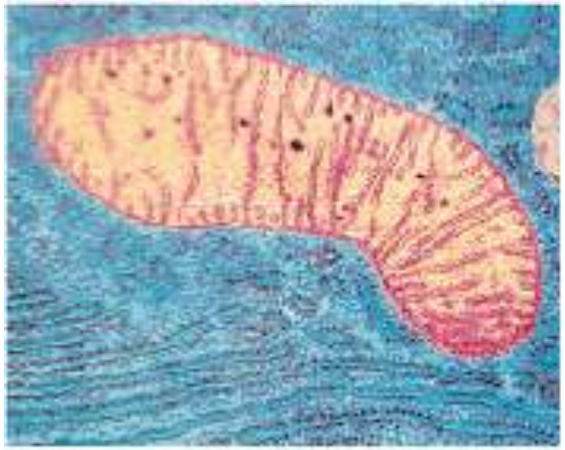
- The processes of electron transport and oxidative phosphorylation are **membrane-associated.**
- In **bacteria**, the conversion of energy from NADH and [$FADH_2$] to the energy of ATP via electron transport and oxidative phosphorylation is carried out at (and across) the **plasma membrane**.
- In **eukaryotic cells**, electron transport and oxidative phosphorylation are localized in **mitochondria**, which are also the sites of TCA cycle activity and fatty acid oxidation.
- Mammalian cells contain from 800 to 2500 mitochondria; other types of cells may have as few as one or two or as many as half a million mitochondria.
  - **Human erythrocytes**, whose purpose is simply to tissues, contain **no mitochondria** at all.
  - The typical mitochondrion is about 0.5 ± 0.3 microns in diameter and from 0.5 micron to several microns long; its overall shape is sensitive to metabolic conditions in the cell.



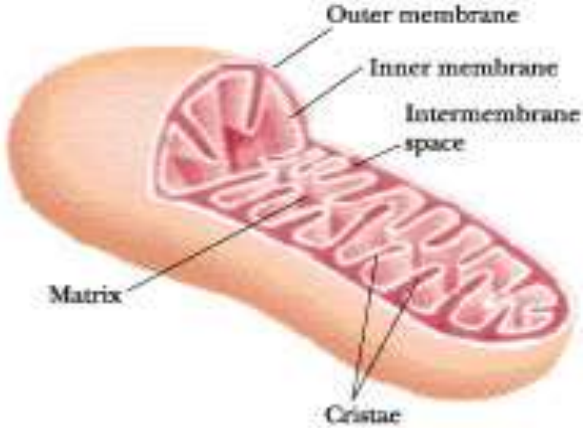
Electron Transport and Oxidative Phosphorylation

Mitochondria

- Mitochondria are surrounded by a simple **outer membrane** and a more complex **inner membrane.**
- The space between the inner and outer membranes is referred to as the **intermembrane space.**
  - Several enzymes that utilize ATP (such as creatine kinase and adenylate kinase) are found in the intermembrane space.
- Apparently, the **outer membrane** functions mainly to maintain the **shape of the mitochondrion**. The inner membrane is richly packed with proteins and its density is higher than that of the outer membrane. The inner membrane lacks cholesterol and is quite impermeable to molecules and ions.
  - specific transport proteins in the membrane carry the ions, substrates, fatty acids for oxidation, and so on across the membrane



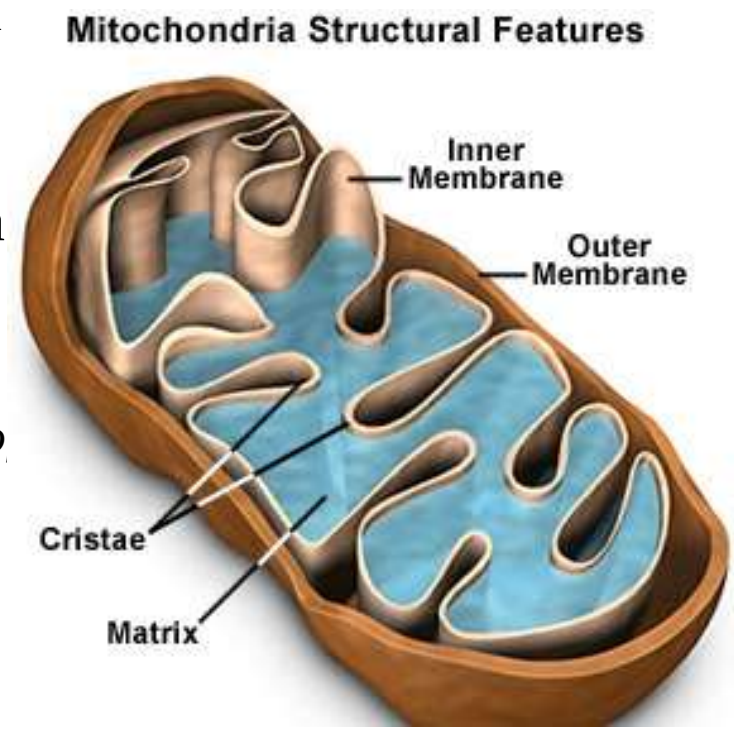
(a)



(b)

(a) An electron micrograph of a mitochondrion. (b) A drawing of a mitochondrion with components labelled.

- The inner membrane is extensively folded . The folds, known as **cristae,** provide the inner membrane with a large surface area in a small volume.
  - During periods of active respiration, the inner membrane appears to shrink significantly, leaving a comparatively large intermembrane space.
- The space inside the inner mitochondrial membrane is called the **matrix,** and it contain most of the enzymes of the TCA cycle and fatty acid oxidation.
  - *An important exception, succinate dehydrogenase o the TCA cycle, is located in the inner membrane itself.*
  - In addition, mitochondria contain circular DNA molecules, along with ribosomes and the enzymes required to synthesize proteins coded within the mitochondrial genome.
  - Although some of the mitochondrial proteins are made this way, most are encoded by nuclear DNA and synthesized by cytosolic ribosomes.



Biological Oxidation

- The evolution of photosynthesis, and the generation of the **oxygen** that is now plentiful in our environment, allowed development of **metabolic pathways** that derive **energy** from transfer of electrons from various reductants ultimately to molecular oxygen.
- **Oxidation is defined as the loss of electrons and reduction as the gain of electrons**.

$$\mathbf{Fe^{++}\ (reduced) \rightarrow Fe^{+++}\ (oxidized) + e^{-}}$$

- **Oxidation-Reduction** = transfer of electrons from electron donor to electron acceptor
  - Example: A:H + B = A + B:H

    donor *acceptor*
- electron *donor* (reducing agent, reductant) is itself oxidized
- electron *acceptor* (oxidizing agent, oxidant) is itself reduced
  - both oxidation and reduction must occur simultaneously
  - one compound donates electrons to (reduces) a second compound; the second accepts electrons from (oxidizes) the first.

- **Biological redox** = Two half-reactions

A:H A

Reductant ⇆ *Oxidant* + e^-

B B:H

Oxidant + e^- ⇆ *Reductant*

(acceptor) *(donor)*

- **Standard reduction potential, E°** :measure of the tendency of oxidant to gain electrons, to become reduced, a potential energy.
  - So, the more *negative* the reduction potential is, the easier a reductant can reduce an oxidant and
  - The more *positive* the reductive potential is, the easier an oxidant can oxidize a reductant
  - The ***difference*** in reduction potential must be important

- **Reduction Potential Difference =ΔE°′**

  $\Delta E^{o\prime} = E^\circ$ (acceptor) - E° (donor)

  ♦measured in volts.

  ♦The more *positive* the reduction potential difference is, the easier the redox reaction

  ♦Work can be derived from the transfer of electrons and the ETS can be used to synthesize ATP.

- The reduction potential can be related to free energy change by:

$$\Delta G^{o\prime} = -nF\Delta E^{o\prime}$$

where n = # electrons transferred = 1,2,3

F = 96.5 kJ/volt, called the Faraday constant

Table of Standard Reduction Potentials

Oxidant + e^- ⇆ *reductant*

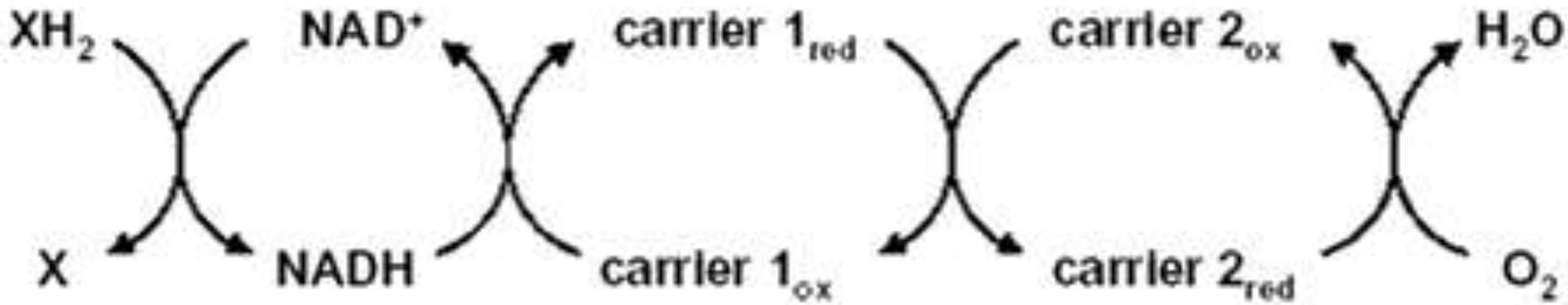
Note:

- *oxidants can oxidize* every compound with less positive voltage -- (below it in the Table)
- *reductants can reduce* every compound with a less negative voltage -- (above it in the Table)

Standard Reduction Potentials

Oxidant	Reductant	n	E°, volt
½O_2+2H^+	H_2O	2	+0.82
oxaloacetate	malate	2	-0.17
pyruvate	lactate	2	-0.19
acetaldehyde	ethanol	2	-0.20
NAD^+	NADH	2	-0.32

- Electrons can move through a chain of donors and acceptors
- In the electron transport chain, electrons flow down a gradient.
- Electrons move from a carrier with low reduction potential (high tendency to donate electrons) toward carriers with higher reduction potential (high tendency to accept electrons).



- **The overall voltage drop from NADH**

  $E°' = -(-0.32\ V)$

  to O **$E°' = +0.82\ V$**

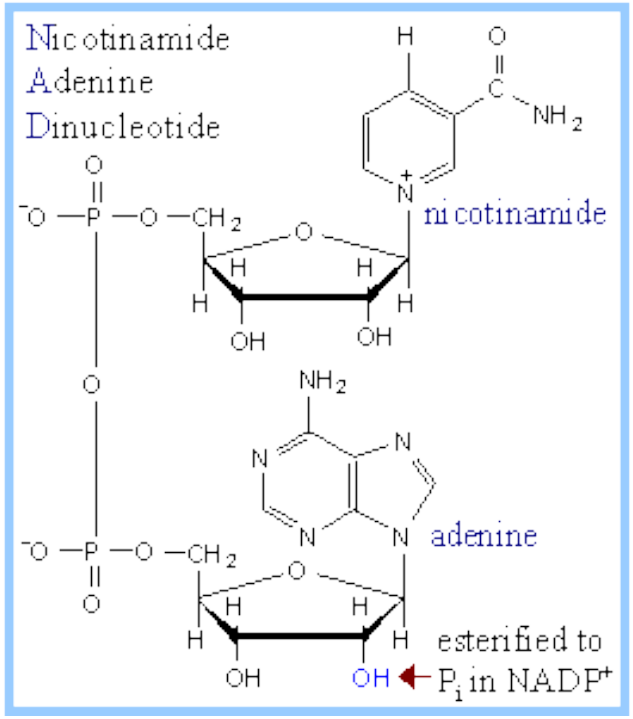
  is **$\Delta E°' = 1.14\ V$**
- This corresponds to **a large free energy change of**

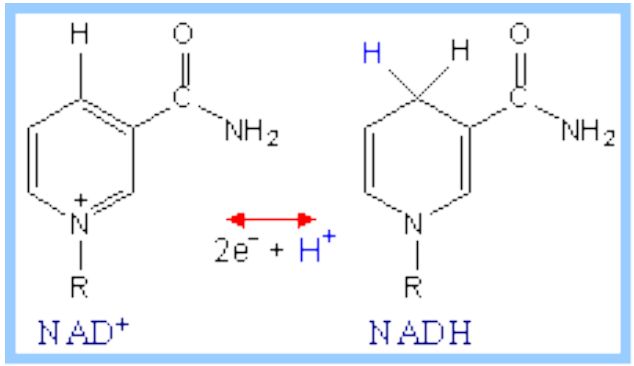
  $\Delta G°' = - nF\Delta E°' =$ -220 kJ/mole (n =2)
-
- **Since ATP requires 30.5 kJ/mole to form from ADP, more than enough energy is available to synthesize 3 ATPs from the oxidation of NADH**.

Electron Carriers

- Two important electron carriers in metabolism are **NAD$^+$** and **FAD**.
- **NAD$^+$ (Nicotinamide Adenine Dinucleotide)** functions as an electron acceptor in catabolic pathways.
- The **nicotinamide** ring of NAD$^+$, which is derived from the vitamin **niacin**, accepts 2 e$^-$ and one H$^+$ (a hydride) in going to the reduced state, as NAD$^+$ becomes **NADH**.
- The electron transfer reaction may be summarized as:

  NAD$^+$ + 2 e$^-$ + 2H$^+$ ↔ NADH + H$^+$



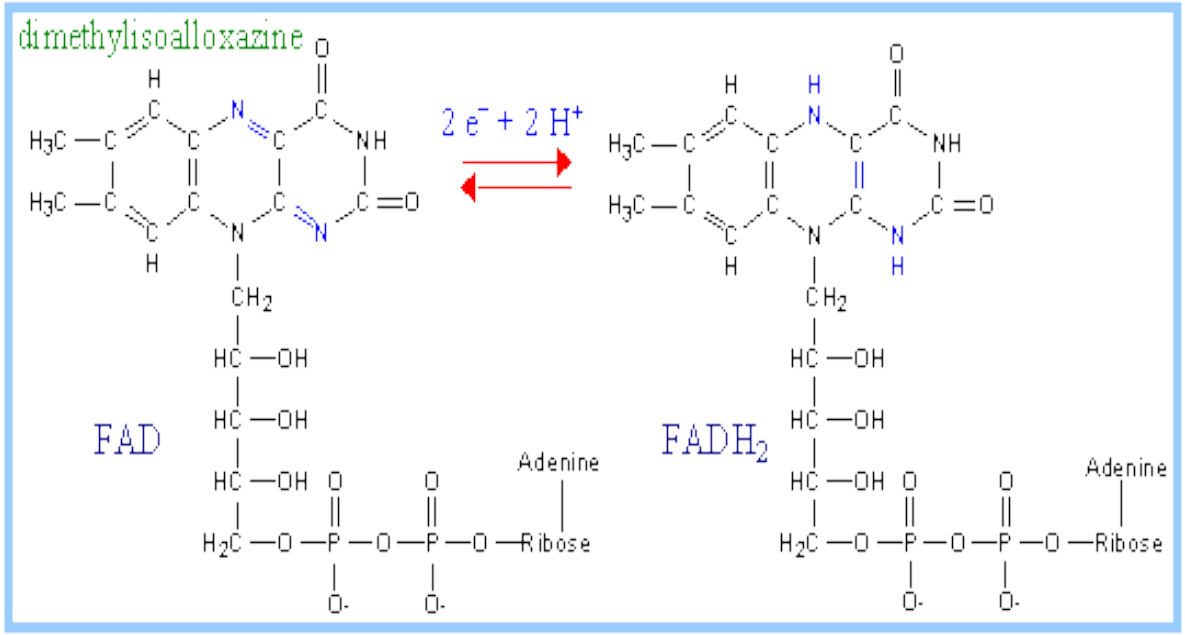


- **NADP$^+$/NADPH** is similar, except for an additional phosphate esterified to a hydroxyl group on the adenosine ribose. NADPH functions as an electron donor in synthetic pathways.

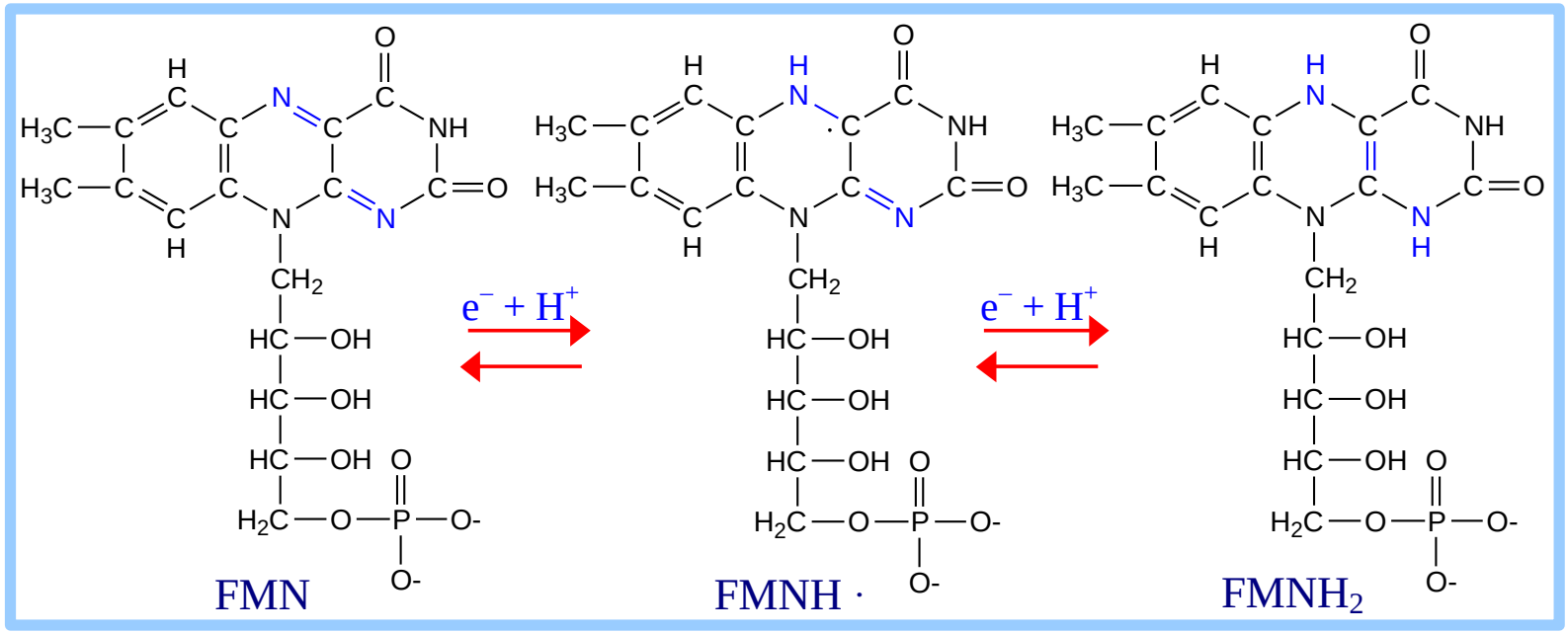
- **<u>FAD (</u>F<u>lavin</u> A<u>denine</u> D<u>inucleotide</u>**) also functions as an electron acceptor. The portion of FAD that undergoes reduction/oxidation is the dimethylisoalloxazine ring, derived from the vitamin **riboflavin**.
- FAD normally accepts 2 e^- and 2 H^+ in going to its reduced state:

$$\mathbf{FAD + 2\,e^- + 2\,H^+ \leftrightarrow FADH_2}$$

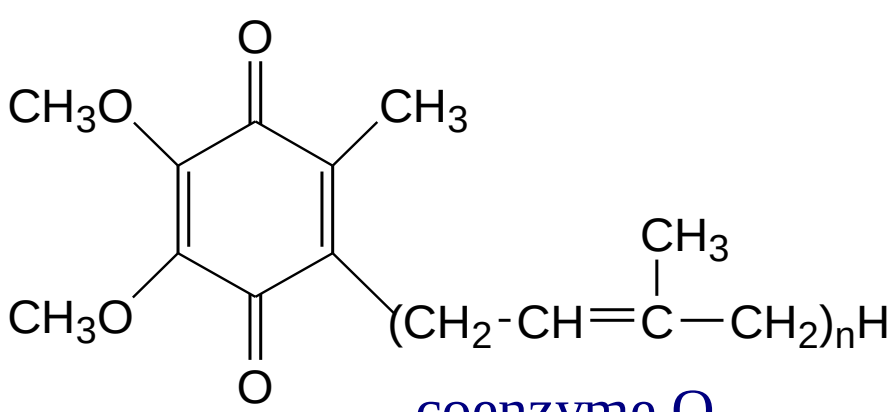
  - NAD^+ is a **coenzyme**, that reversibly binds to enzymes.
  - FAD is a **prosthetic group**, that usually remains tightly bound at the active site of an enzyme.



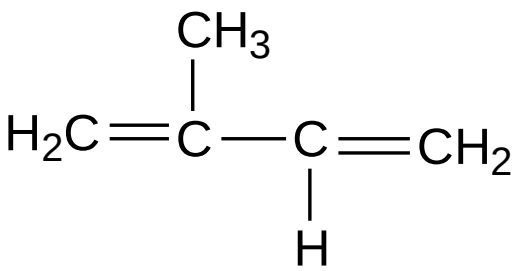
- **<u>FMN (Flavin MonoNucleotide)</u>** is a prosthetic group of some flavoproteins. It is similar in structure to FAD (Flavin Adenine Dinucleotide), but lacking the adenine nucleotide.
- FMN (like FAD) can accept 2 e^- + 2 H^+ to yield $FMNH_2$. When bound at the active site of some enzymes, FMN can accept 1 e^-, converting it to the half-reduced **semiquinone radical**. The semiquinone can accept a second e^- to yield $FMNH_2$.
- **Role of FMN:** Since it can accept/donate either **1 or 2 e^-**, FMN has an important role in mediating electron transfer between carriers that transfer 2 e^- (e.g., NADH) and carriers that can only accept 1 e^- (e.g., Fe^{+++}).



- **Coenzyme Q** (CoQ, Q, ubiquinone) is very **hydrophobic**.
  It dissolves in the hydrocarbon core of a membrane.
- It includes a long **isoprenoid tail**, with multiple units having a carbon skeleton comparable to that of isoprene.
  In human cells, most often n = 10.
- **Q_{10}'s isoprenoid tail** is longer than the width of a bilayer.
  It may be folded to yield a more compact structure, & is postulated to reside in the central domain of a membrane, **between the 2 lipid monolayers**.
- **Coenzyme Q** functions as a mobile e^- carrier within the mitochondrial inner membrane.
- It plays a role in trans-membrane **H^+ transport** coupled to e^- transfer (**Q Cycle**)

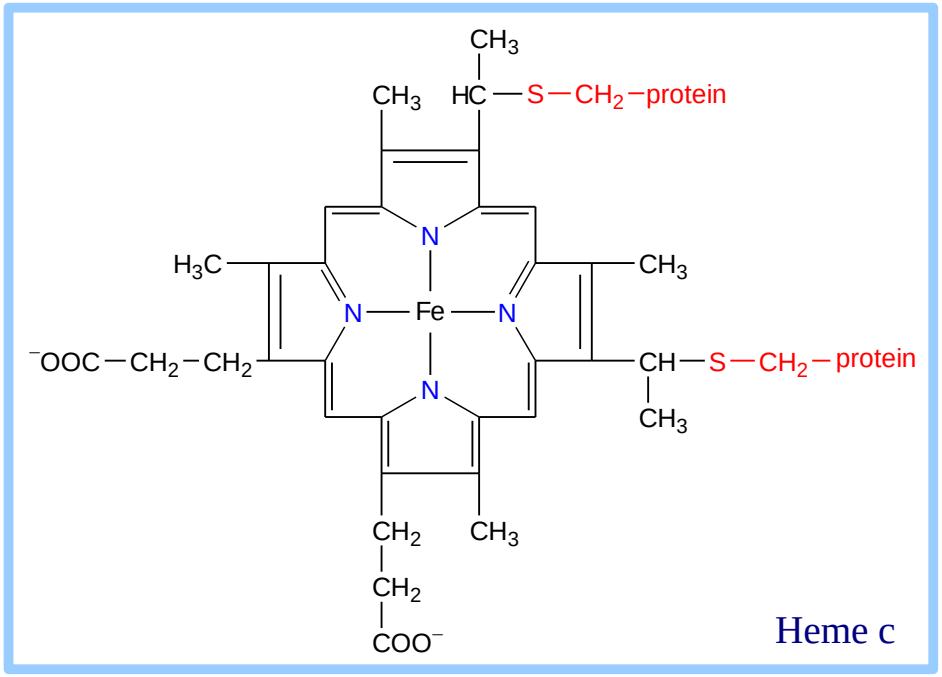


coenzyme Q



isoprene

- **Heme** is a prosthetic group of **cytochromes**.
- Heme contains an iron atom in a porphyrin ring system. The **Fe** is bonded to 4 **N** atoms of the porphyrin ring.
- Hemes in the 3 classes of cytochrome (**a**, **b**, **c**) differ slightly in substituents on the porphyrin ring system. A common feature is 2 propionate side-chains. Only **heme c** is covalently linked to the protein via thioether bonds to **cysteine** residues.
- The heme iron can undergo a **1 e⁻** transition between ferric and ferrous states: $Fe^{+++} + e^- \leftrightarrow Fe^{++}$



•**Cytochromes** are proteins with heme prosthetic groups. They absorb light at characteristic wavelengths.

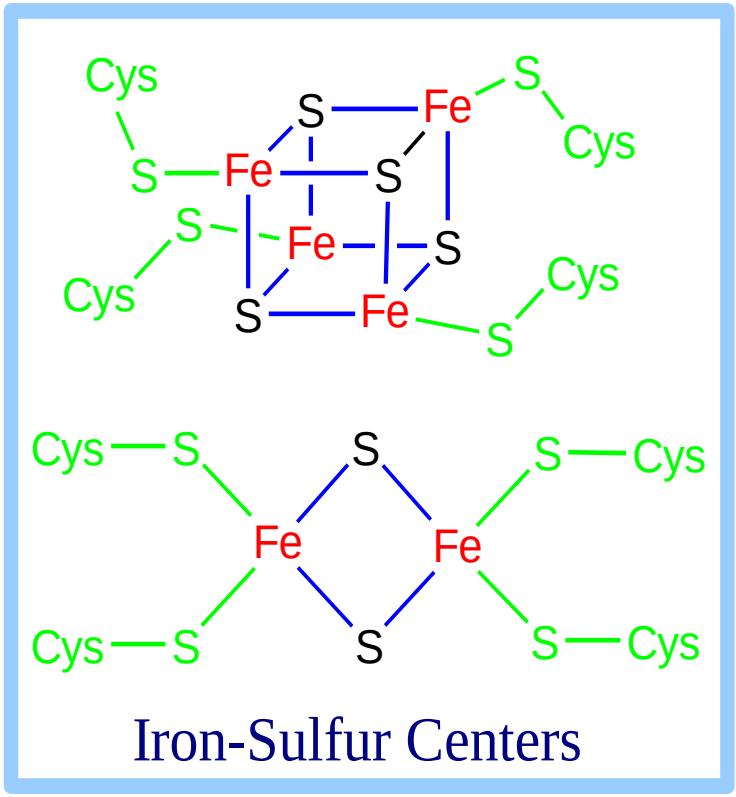
–**Absorbance changes** upon oxidation/reduction of the heme iron provide a basis for monitoring heme redox state.

- For the hemoglobin heme group:
  a) Fe has *his* bound from above and O_2, comes in below to bind.
  b) Fe remains as Fe(II) & no electrons are transferred.
  – Just carries O_2

- For *cytochrome heme,*
  a) all 6 sites of Fe are filled (4 from porphyrin, 2 above and below from protein) = no *molecule* can approach
  b) carries electrons only: $Fe(III) + e^- \leftrightarrows Fe(II)$
- Note: Only *one* electron is transferred at a time.

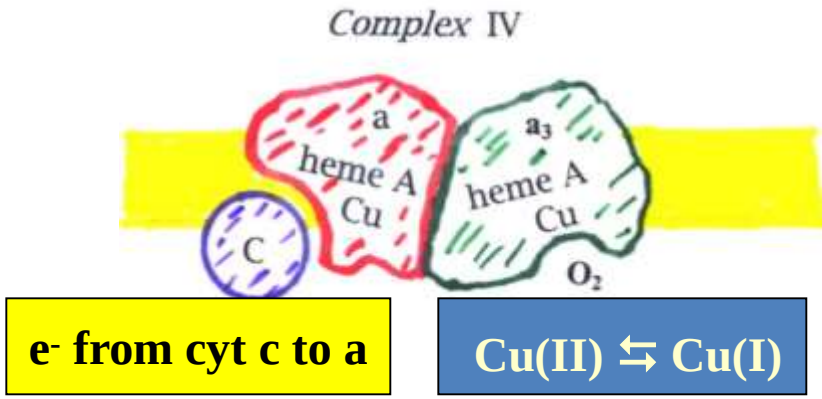
- Cytochromes in respiration are on *inner mitochondrial membrane*
- ☛ ***cytochromes b, c_1, c, a, a_3***, relay electrons, one at a time, in this order

- A number of **iron-sulfur proteins,** which participate in one-electron transfers involving the Fe^{2+} and Fe^{3+} states.
  - Transfer of electrons in variety of proteins
- **Cysteine** residues provide **S** ligands to the iron, while also holding these prosthetic groups in place within the protein.



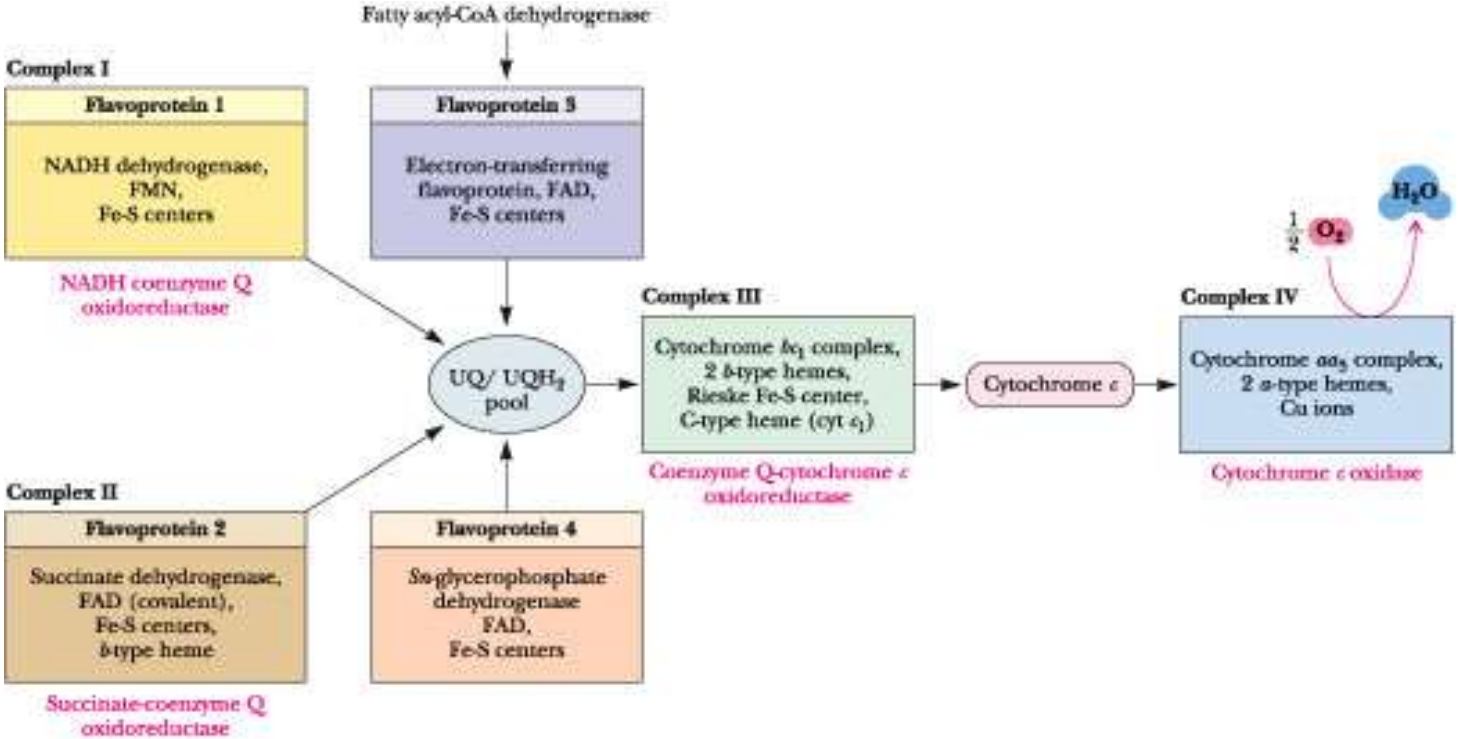
Iron-Sulfur Centers

- Protein-bound **copper,** a one-electron transfer site, which converts between Cu^{-} and Cu^{2+}.
  - Heme A and Cu act together to transfer electrons to oxygen

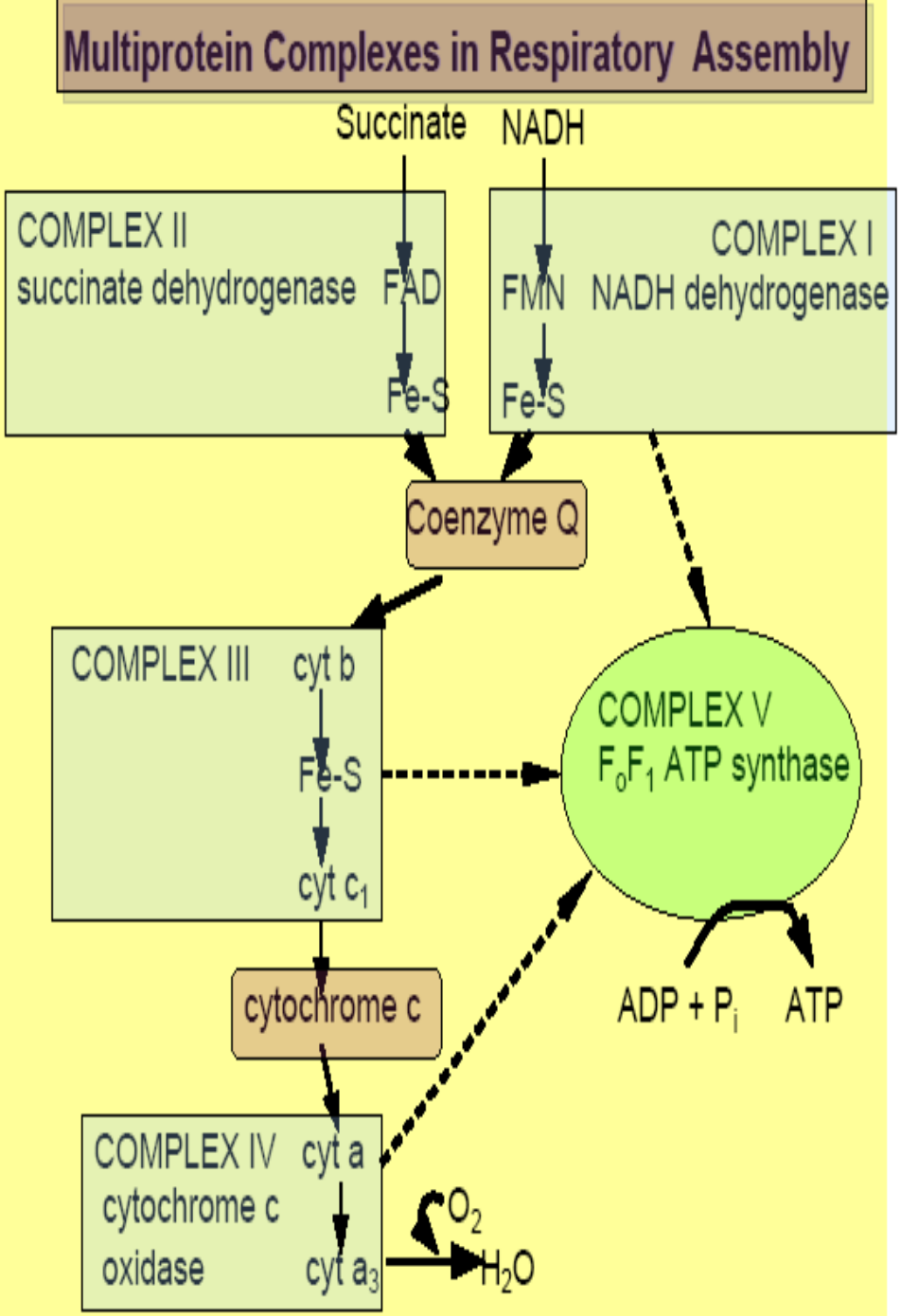


An overview of the complexes and pathways in the mitochondrial electron transport chain

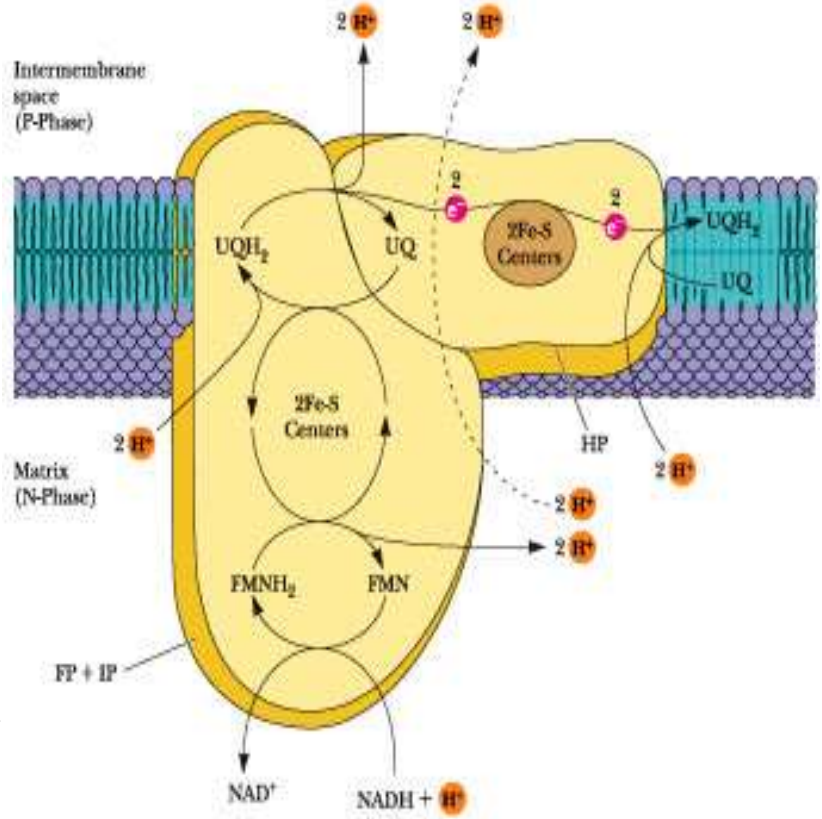
- The complete chain can be considered to be composed of four parts: (I) **NADH-coenzyme Q reductase,** (II) **succinate-coenzyme Q reductase,** (III) **coenzyme Q-cytochrome *c* reductase,** and (IV) **cytochrome *c* oxidase**
- **Complex I** accepts electrons from NADH, serving as a link between glycolysis, the TCA cycle, fatty acid oxidation, and the electron transport chain.
- **Complex II** includes succinate dehydrogenase and thus forms a direct link between the TCA cycle and electron transport.



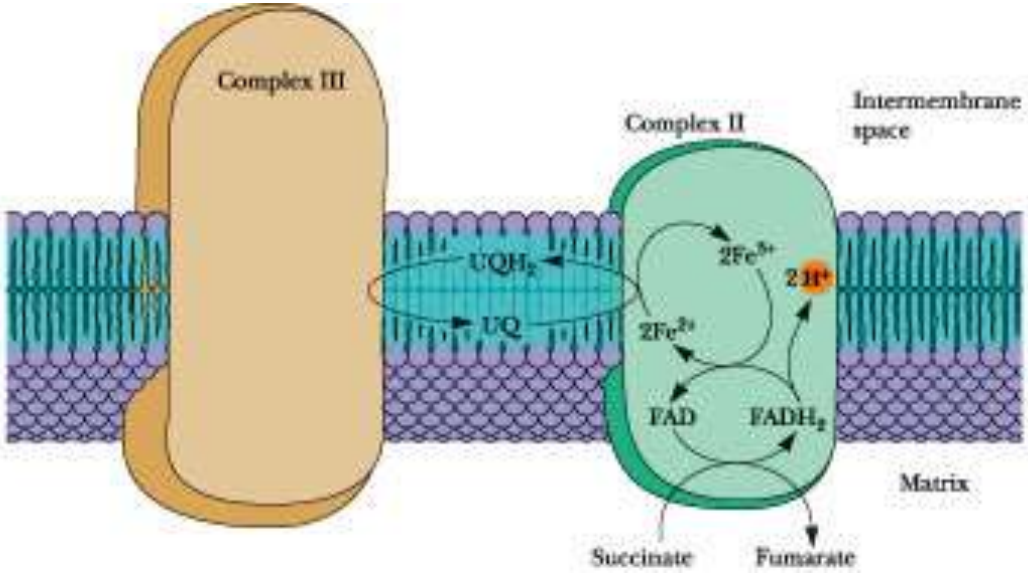
- Complexes I and II produce a common product, reduced coenzyme Q (UQH_2), which is the substrate for coenzyme Q-cytochrome *c* reductase (Complex III).
- There are two other ways to feed electrons to UQ: the **electron-transferring flavoprotein,** which transfers electrons from the flavoprotein-linked step of fatty acyl-CoA dehydrogenase, and ***sn*-glycerophosphate dehydrogenase.**
- **Complex III** oxidizes UQH_2 while reducing cytochrome *c*, which in turn is the substrate for Complex IV, cytochrome *c* oxidase.
- **Complex IV** is responsible for reducing molecular oxygen.



- **Complex I: NADH-Coenzyme Q Reductase**
  - This complex transfers a pair of electrons from NADH to coenzyme Q, a small, hydrophobic, yellow compound.
  - Another common name for this enzyme complex is *NADH dehydrogenase*.
  - the first step is binding of NADH to the enzyme on the matrix side of the inner mitochondrial membrane, and transfer of electrons from NADH to tightly bound FMN
  - The second step is the transfer of electrons from the reduced [$FMNH_2$] to a series of Fe-S proteins
  - The final step is the transfer of two electrons from iron-sulfur clusters to coenzyme Q
  - The oxidation of one NADH and the reduction of one UQ by NADH-UQ reductase results in the net transport of protons from the matrix side to the cytosolic side of the inner membrane.

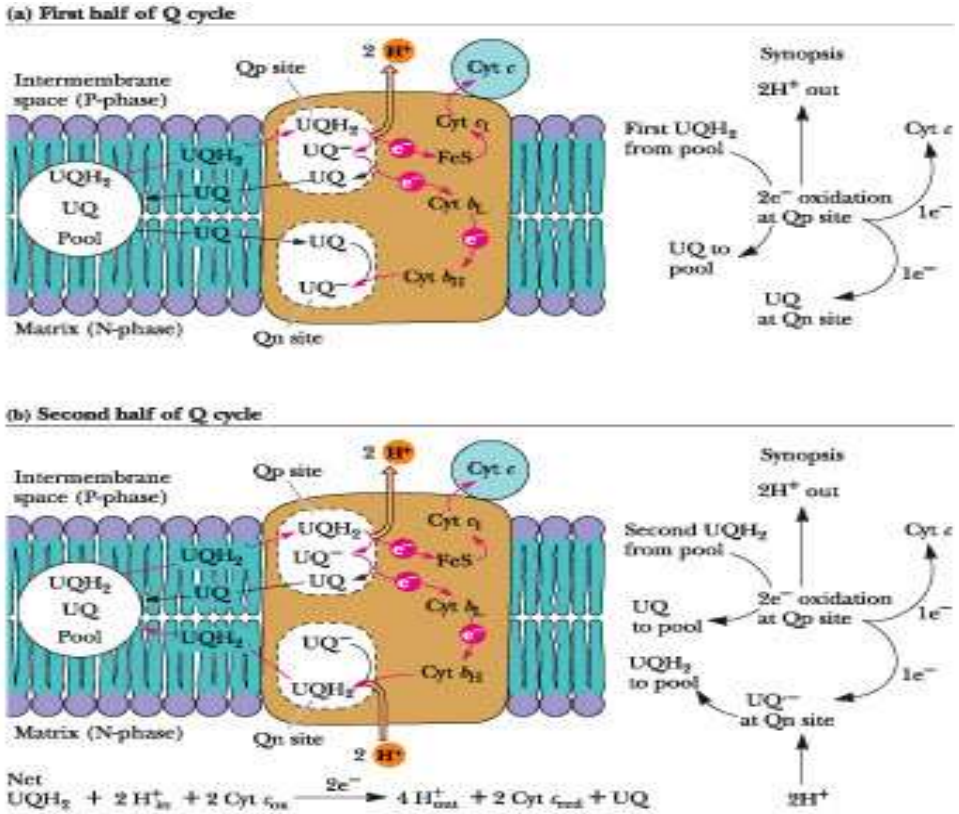


- **Complex II: Succinate-Coenzyme Q Reductase**
  - Complex II is perhaps better known as **succinate dehydrogenase,** the only TCA cycle enzyme that is an integral membrane protein in the inner mitochondrial membrane.
  - When succinate is converted to fumarate in the TCA cycle, concomitant reduction of bound FAD to $FADH_2$ occurs in succinate dehydrogenase.
  - This $FADH_2$ transfers its electrons immediately to Fe-S centers, which pass them on to UQ.
  - Other enzymes can also supply electrons to UQ, including mitochondrial *sn*-glycerophosphate dehydrogenase, an inner membrane-bound shuttle enzyme, and the fatty acyl-CoA dehydrogenases, three soluble matrix enzymes involved in fatty acid oxidation



A probable scheme for electron flow in Complex II. Oxidation of succinate occurs with reduction of [FAD]. Electrons are then passed to Fe-S centers and then to coenzyme Q (UQ). Proton transport does not occur in this complex.

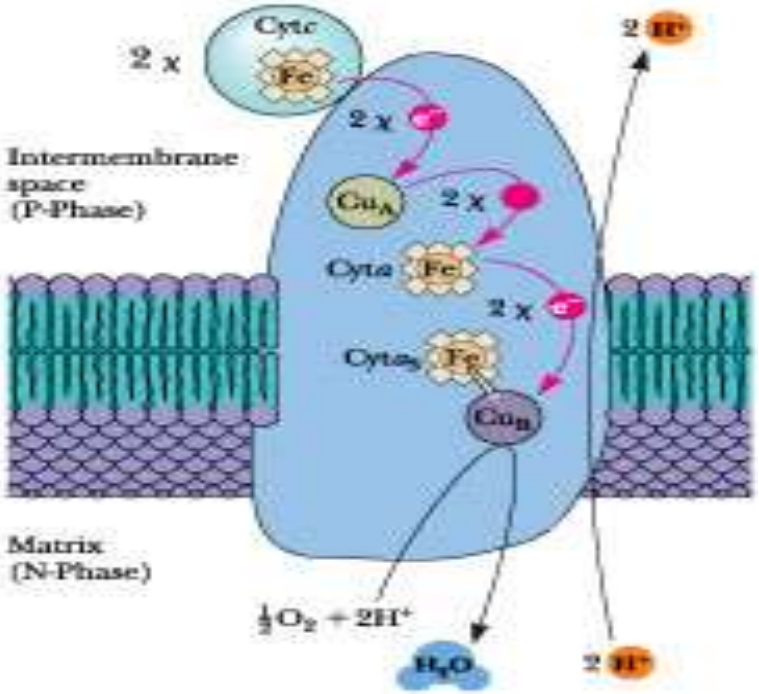
- **Complex III: Coenzyme Q-Cytochrome *c* Reductase**
  - In the third complex of the electron transport chain, reduced coenzyme Q (UQH_2) passes its electrons to cytochrome *c* via a unique redox pathway known as the **Q cycle.**
  - As with Complex I, passage of electrons through the Q cycle of Complex III is accompanied by proton transport across the inner mitochondrial membrane.
  - A large pool of UQ and UQH_2 exists in the inner mitochondrial membrane. The Q cycle is initiated when a molecule of UQH_2 from this pool diffuses to a site (called $\mathbf{Q_p}$) on Complex III near the cytosolic face of the membrane.



- Complex III takes up two protons on the matrix side of the inner membrane and releases four protons on the cytoplasmic side for each pair of electrons that passes through the Q cycle.
- The apparent imbalance of *two protons in* for *four protons out* is offset by proton translocations in Complex IV, the cytochrome oxidase complex.

- **Complex IV: Cytochrome *c* Oxidase**
  - Complex IV is called cytochrome *c* oxidase because it accepts electrons from cytochrome *c* and directs them to the four-electron reduction of O_2 to form H_2O:

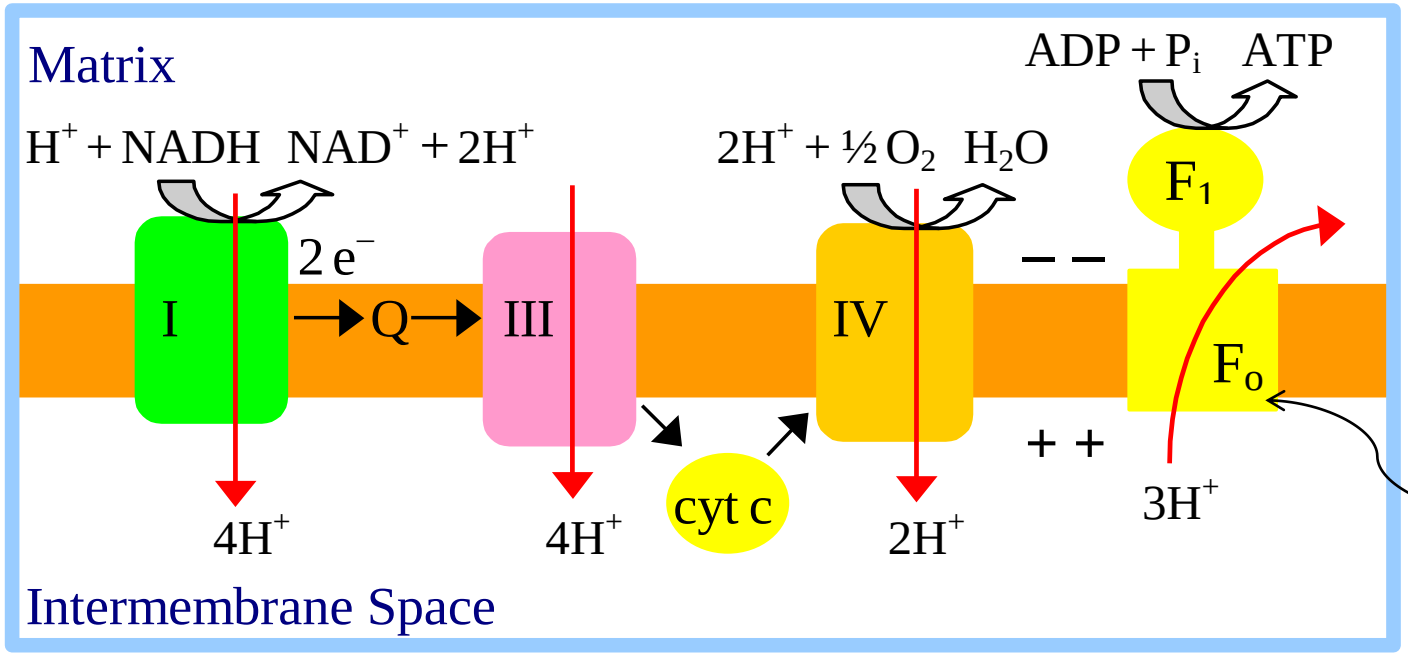
    $4\ \text{cyt}\ c\ (Fe^{2+}) + 4\ H^+ + O_2 \rightarrow 4\ \text{cyt}\ c\ (Fe^{3+}) + 2\ H_2O$
  - Thus, O_2 and cytochrome *c* oxidase are the final destination for the electrons derived from the oxidation of food materials.
  - The reduction of oxygen in Complex IV is accompanied by transport of protons across the inner mitochondrial membrane.
  - Four protons are taken up on the matrix side for every two protons transported to the cytoplasm



The electron transfer pathway for cytochrome oxidase. Cytochrome *c* binds on the cytosolic side, transferring electrons through the copper and heme centers to reduce O_2 on the matrix side of the membrane.

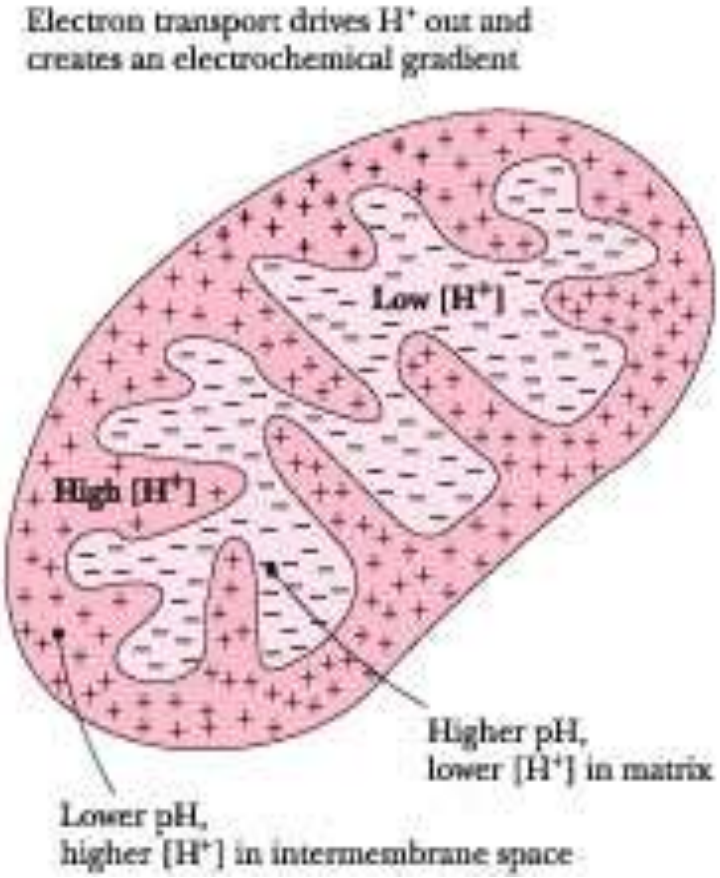
Oxidative Phosphorylation (ox-phos) and Chemiosmotic theory

- Definition: Production of ATP using transfer of electrons for energy
- For many years, the means by which electron transport and ATP synthesis are coupled was unknown.
- E. C. Slater in 1953, proposed that energy derived from electron transport was stored in a **high-energy intermediate** (symbolized as X~P).
  - This chemical species—in essence an activated form of phosphate—functioned according to certain relations to drive ATP synthesis.
- In 1961, Peter Mitchell proposed a novel coupling mechanism involving a proton gradient across the inner mitochondrial membrane.



ATP Synthase: The mitochondrial complex that carries out ATP synthesis is called **ATP synthase** or sometimes **F_1F_0-ATPase**

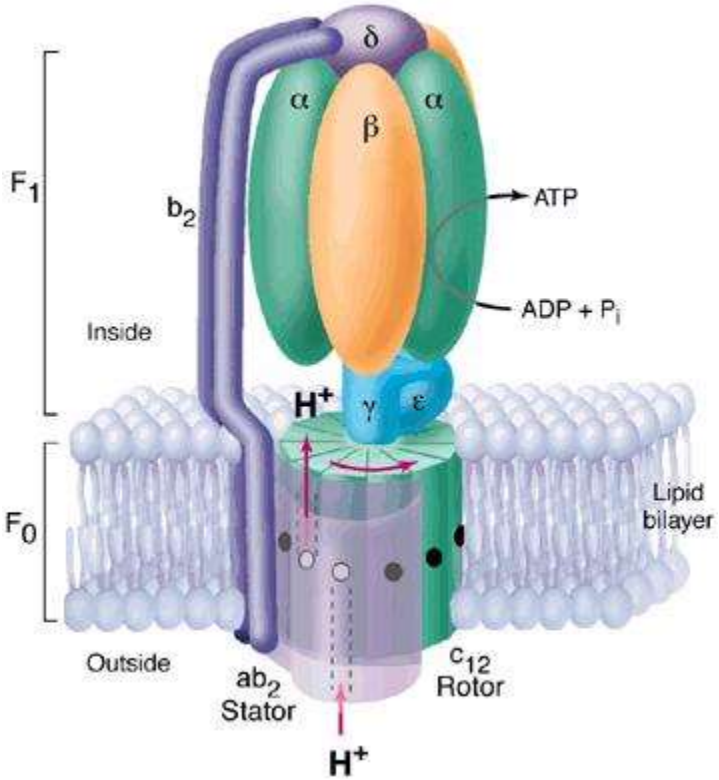
- In Mitchell's **chemiosmotic hypothesis,** protons are driven across the membrane from the matrix to the intermembrane space and cytosol by the events of electron transport.
- This mechanism stores the energy of electron transport in an **electrochemical potential.**
  - As protons are driven out of the matrix, the pH rises and the matrix becomes negatively charged with respect to the cytosol .
  - Proton pumping thus creates a pH gradient and an electrical gradient across the inner membrane, both of which tend to attract protons back into the matrix from the cytoplasm.
  - Flow of protons down this electrochemical gradient, an energetically favorable process, then drives the synthesis of ATP.



The proton and electrochemical gradients existing across the inner mitochondrial membrane. The electrochemical gradient is generated by the transport of protons across the membrane.

Chemiosmotic theory - respiration:

- Spontaneous e^- transfer through complexes I, III, & IV is coupled to non-spontaneous H^+ ejection from the matrix.
- H^+ ejection creates a **membrane potential** ($\Delta\Psi$, negative in matrix) and a **pH gradient** (ΔpH, alkaline in matrix).
- Non-spontaneous ATP synthesis is coupled to spontaneous H^+ transport into the matrix. The pH & electrical gradients created by respiration are the driving force for H^+ uptake.
- H^+ return to the matrix via F_o "uses up" pH & electrical gradients.
- **ATP** produced in the mitochondrial matrix must exit to the **cytosol** to be used by transport pumps, kinases, etc.
  - **ADP** & **P_i** arising from ATP hydrolysis in the cytosol must reenter the matrix to be converted again to ATP.
  - Two carrier proteins in the inner mitochondrial membrane are required.



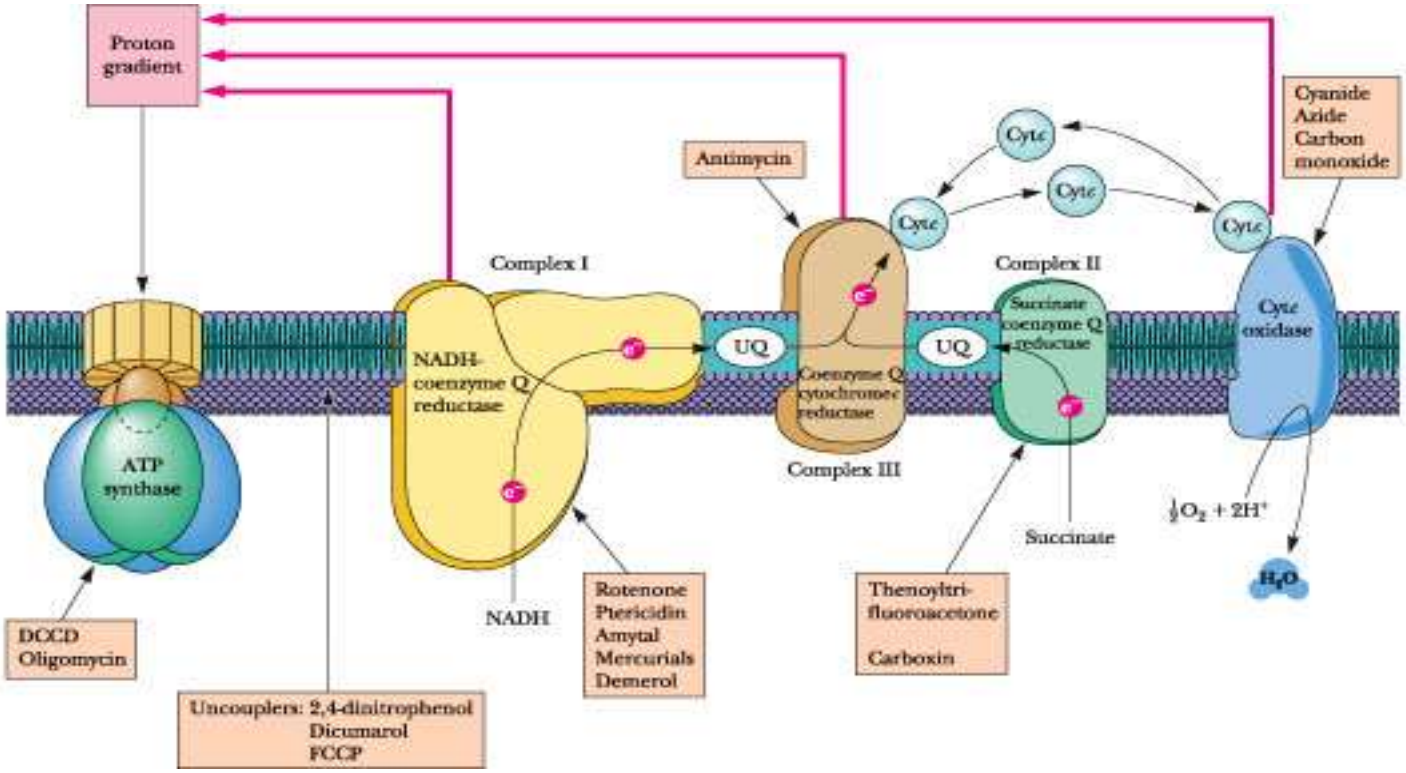
- ETC and oxidative phosphorylation
- http://www.rpi.edu/dept/bcbp/molbiochem/MBWeb/mb1/part2/oxphos.htm#animat2

- http://vcell.ndsu.edu/animations/etc/movie-flash.htm

Control of Respiration

- Most mitochondria are said to be *tightly coupled.*
  - That is there is no electron flow without phosphorylation and no phosphorylation without electron flow.
- Reduced substrate, ADP, P_i and O_2 are all necessary for oxidative phosphorylation.
- For example, in the absence of ADP or O_2 electron flow stops, reduced substrate is not consumed and no ATP is made.
- Brown adipose (fat) cells contain natural uncouplers (thermogenin) to warm animals - cold adaptation and hibernation
  - The uncoupling protein blocks development of a H^+ electrochemical gradient, thereby stimulating respiration. ΔG of respiration is dissipated as heat.
  - This "**non-shivering thermogenesis**" is costly in terms of respiratory energy unavailable for ATP synthesis, but provides valuable warming of the organism.

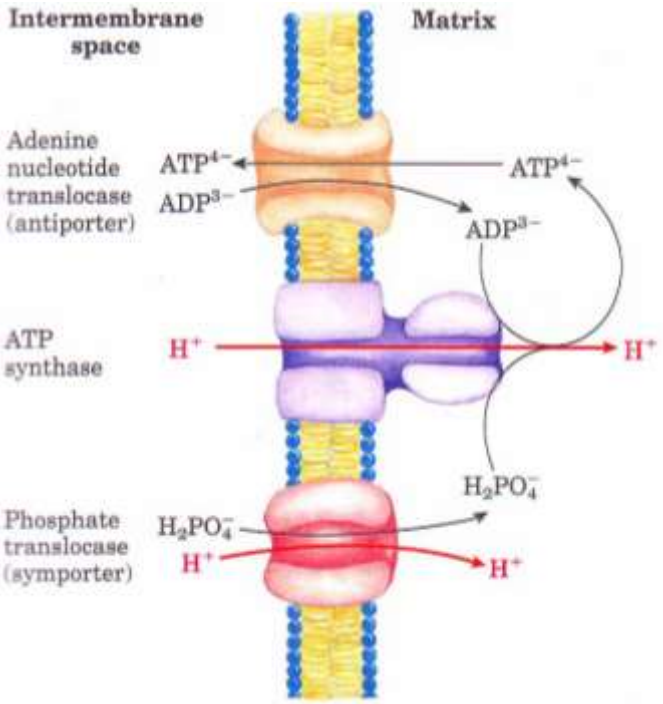
Inhibitors of Oxidative Phosphorylation



- The unique properties and actions of an inhibitory substance can often help to identify aspects of an enzyme mechanism.
- Many details of electron transport and oxidative phosphorylation mechanisms have been gained from studying the effects of particular inhibitors.
- The sites of inhibition by these agents are indicated in Figure

ATP Exits the Mitochondria via an ATP-ADP Translocase

- ATP, the cellular energy currency, must exit the mitochondria to carry energy throughout the cell, and ADP must be brought into the mitochondria for reprocessing.
- Neither of these processes occurs spontaneously because the highly charged ATP and ADP molecules do not readily cross biological membranes.
- Instead, these processes are mediated by a single transport system, the protein, **ATP-ADP translocase.**
  - For each ATP transported out, one ADP is transported into the matrix
  - It binds ATP on the matrix side, reorients to face the cytosol, and exchanges ATP for ADP, with subsequent movement back to the matrix face of the inner membrane.



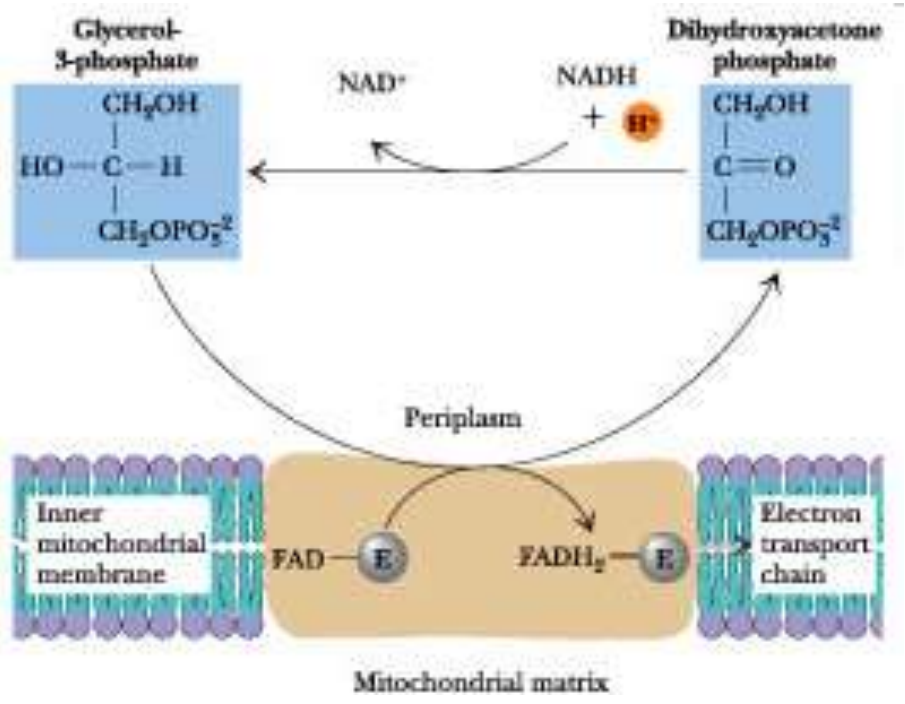
- Approximately 4 protons are transported into the matrix /ATP synthesized.
- Thus, approximately ¼ **of the energy** derived from the respiratory chain (electron transport and oxidative phosphorylation) **is expended** to mitochondrial ATP-ADP transport.

Shuttle pathways – Glycerol phosphate Shuttle, Malate aspartate Shuttle

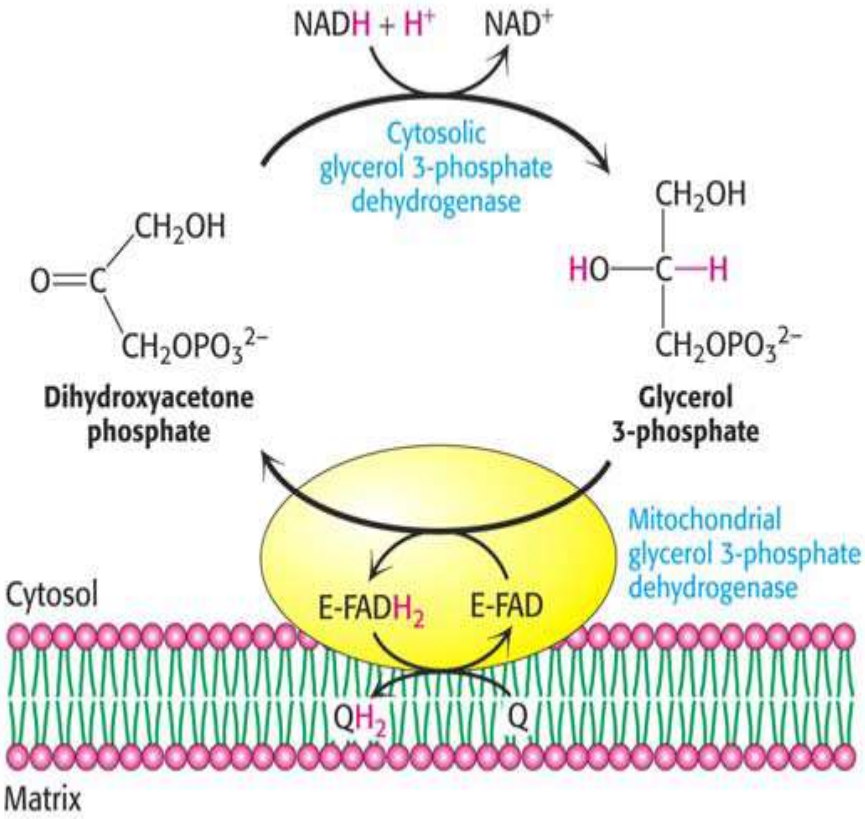
- Most of the NADH used in electron transport is produced in the mitochondrial matrix space, an appropriate site because NADH is oxidized by Complex I on the matrix side of the inner membrane.
- Furthermore, the inner mitochondrial membrane is impermeable to NADH.
- However, that NADH is produced in glycolysis by glyceraldehyde-3-P dehydrogenase in the cytosol.
- If this NADH were not oxidized to regenerate NAD^+, the glycolytic pathway would cease to function due to NAD^+ limitation.
- Eukaryotic cells have a number of *shuttle systems* that harvest the electrons of cytosolic NADH for delivery to mitochondria without actually transporting NADH across the inner membrane

Shuttle pathway – Glycerol phosphate (glycerophosphate) Shuttle

- In the **glycerophosphate shuttle,** two different **glycerophosphate dehydrogenases,** one in the cytoplasm and one on the outer face of the mitochondrial inner membrane, work together to carry electrons into the mitochondrial matrix
- NADH produced in the cytosol transfers its electrons to *dihydroxyacetone phosphate*, thus reducing it to *glycerol-3-phosphate.*
- This metabolite is reoxidized by the FAD^-dependent mitochondrial membrane enzyme to reform dihydroxyacetone phosphate and enzyme-bound $FADH_2$.

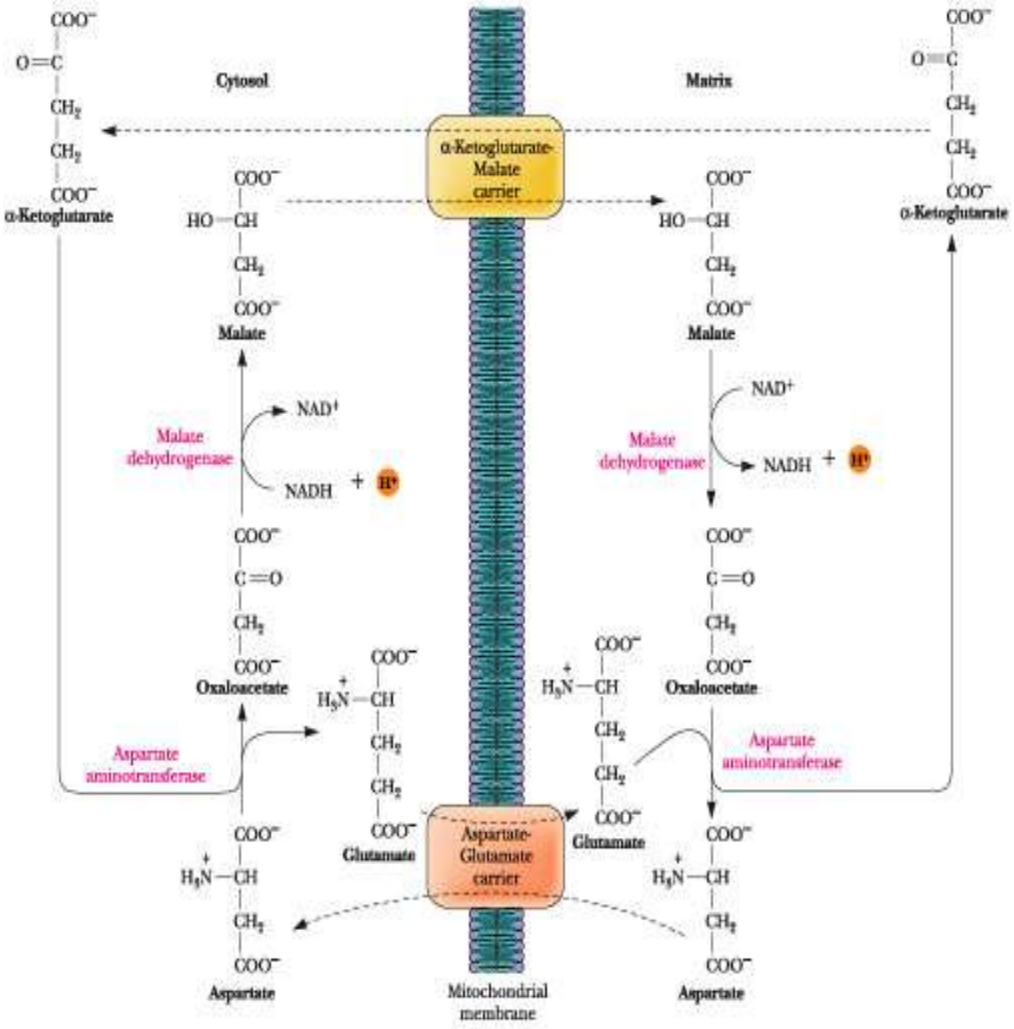


- The two electrons of [$FADH_2$] are passed directly to UQ, forming UQH_2.
- Thus, via this shuttle, cytosolic NADH can be used to produce mitochondrial [$FADH_2$] and, subsequently, UQH_2. As a result, cytosolic NADH oxidized via this shuttle route yields only 1.5 molecules of ATP.
- The cell "pays" with a potential ATP molecule for the convenience of getting cytosolic NADH into the mitochondria.
- Although this may seem wasteful, there is an important payoff. The glycerophosphate shuttle is essentially irreversible, and even when NADH levels are very low relative to NAD^+, the cycle operates effectively.

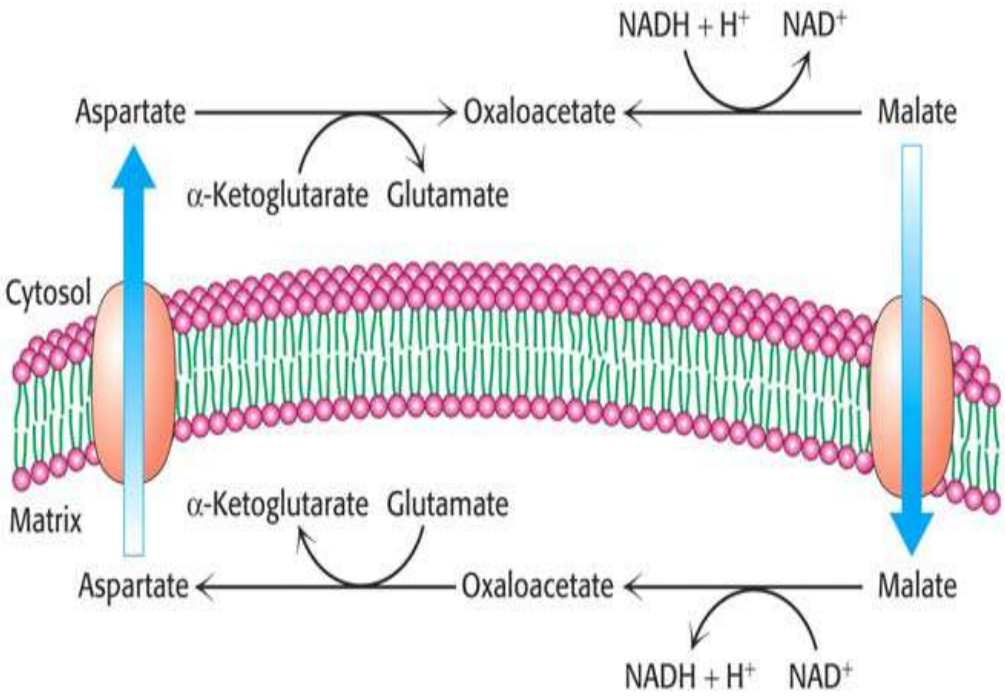


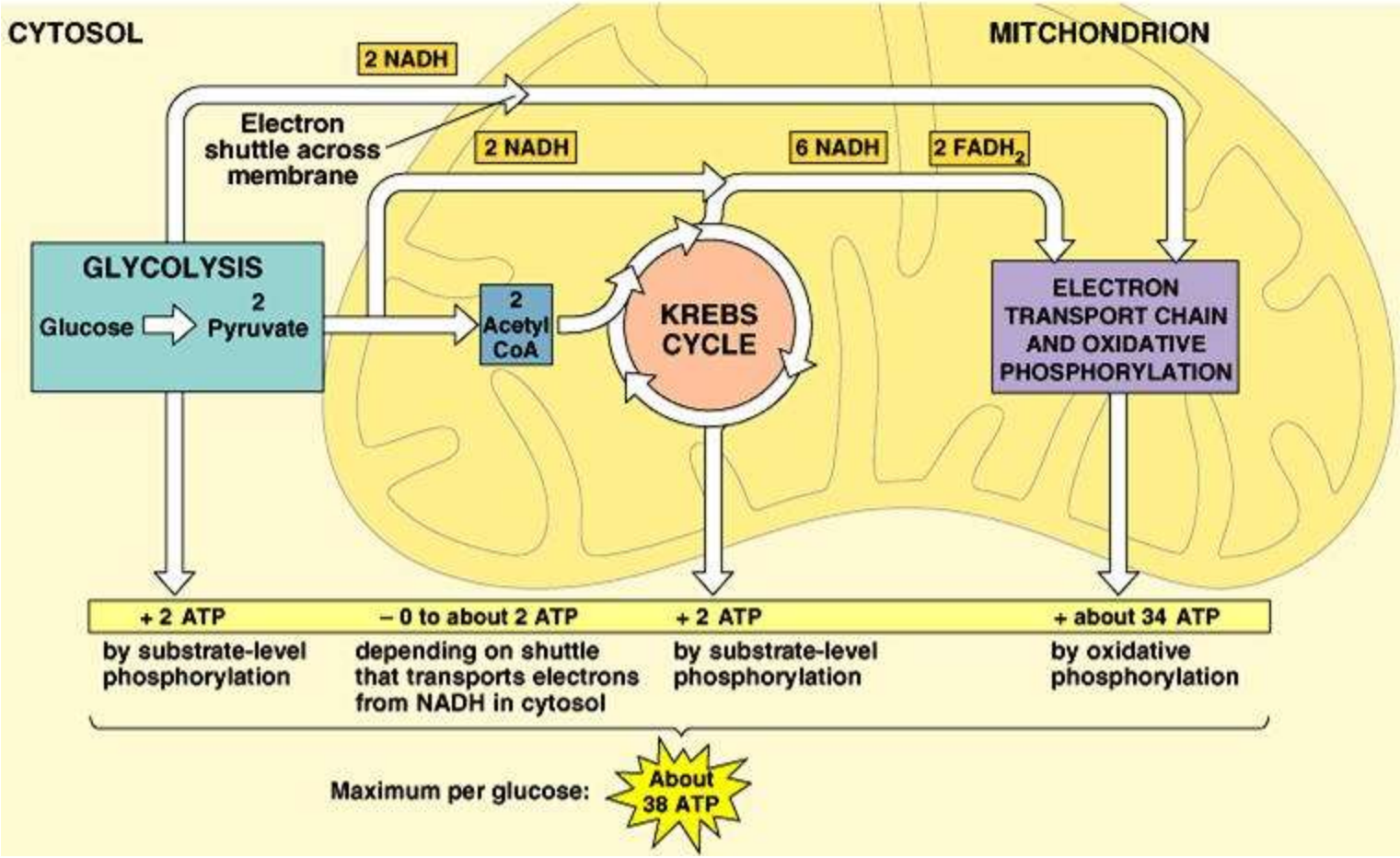
Shuttle pathways –Malate aspartate Shuttle

- The second electron shuttle system is the **malate-aspartate shuttle**
- Oxaloacetate is reduced in the cytosol, acquiring the electrons of NADH (which is oxidized to NAD^+).
- Malate is transported across the inner membrane, where it is reoxidized by malate dehydrogenase, converting NAD^+ to NADH in the matrix.
- This mitochondrial NADH readily enters the electron transport chain.



- The oxaloacetate produced in this reaction cannot cross the inner membrane and must be transaminated to form *aspartate, which can be transported* across the membrane to the cytosolic side
- Transamination in the cytosol recycles aspartate back to oxaloacetate.
- In contrast to the glycerol phosphate shuttle, **the malate-aspartate cycle is reversible**
- It operates only if the NADH/NAD$^+$ ratio in the cytosol is higher than the ratio in the matrix. Because this shuttle produces NADH in the matrix, the full 2.5 ATPs per NADH are recovered.





- **Prokaryotic cells** need not carry out ATP/ADP exchange.
- Thus, bacteria have the potential to produce approximately 38 ATP per glucose.

Net Yield of ATP from Glucose Oxidation Depends on the Shuttle Used

- In **eukaryotic cells**, the combined pathways of glycolysis, the TCA cycle, electron transport, and oxidative phosphorylation then yield a net of approximately 30 to 32 molecules of ATP per molecule of glucose oxidized, depending on the shuttle route employed
- These P/O ratios of 2.5 and 1.5 for mitochondrial oxidation of NADH and [$FADH_2$] are "**consensus values**."
- These do not reflect actual values and because these ratios may change depending on metabolic conditions, **these estimates of ATP yield from glucose oxidation are approximate.**

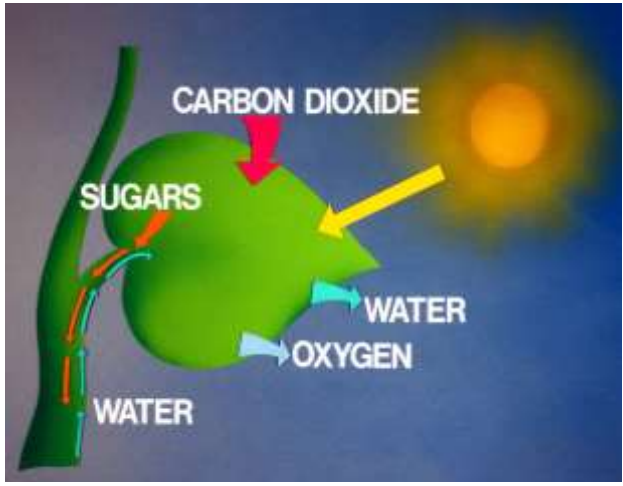
	ATP Yield per Glucose	
	Glycerol-Phosphate Shuttle	Malate-Aspartate Shuttle
Glycolysis: glucose to pyruvate (cytosol)		
Phosphorylation of glucose	-1	-1
Phosphorylation of fructose-6-phosphate	-1	-1
Dephosphorylation of 2 molecules of 1,3-BPG	+2	+2
Dephosphorylation of 2 molecules of PEP	+2	+2
Oxidation of 2 molecules of glyceraldehyde-3-phosphate yields 2 NADH		
Pyruvate conversion to acetyl-CoA (mitochondria)		
2 NADH		
Citric acid cycle (mitochondria)		
2 molecules of GTP from 2 molecules of succinyl-CoA	+2	+2
Oxidation of 2 molecules each of isocitrate, a-ketoglutarate, and malate yields 6 NADH		
Oxidation of 2 molecules of succinate yields 2 [$FADH_2$]		
Oxidative phosphorylation (mitochondria)		
2 NADH from glycolysis yield 1.5 ATP each if NADH is oxidized by glycerol-phosphate shuttle; 2.5 ATP by malate-aspartate shuttle	+3	+5
Oxidative decarboxylation of 2 pyruvate to 2 acetyl-CoA: 2 NADH produce 2.5 ATP each	+5	+5
2 [$FADH_2$] from each citric acid cycle produce 1.5 ATP each	+3	+3
6 NADH from citric acid cycle produce 2.5 ATP each	+15	+15
Net Yield	30	32

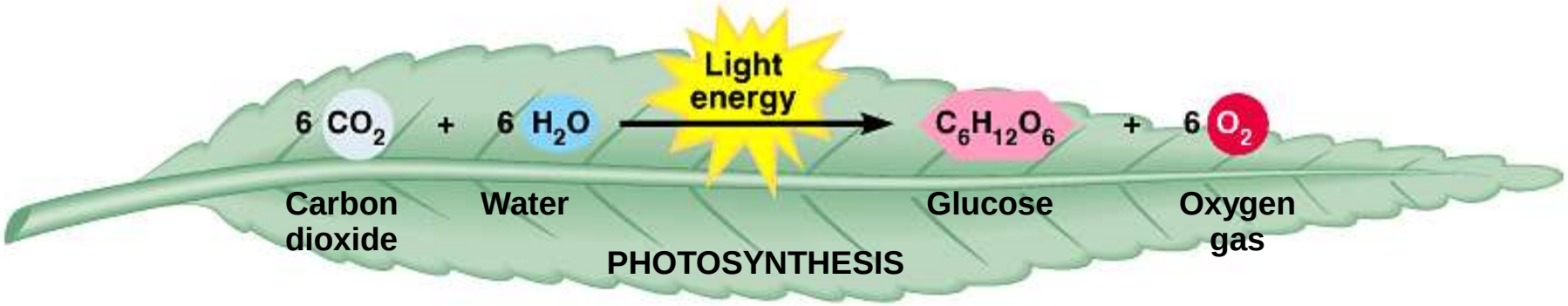
How much energy does a eukaryotic cell extract from the glucose molecule?

- Glucose + $6O_2 \rightarrow 6CO_2 + 6H_2O$
- $\Delta G = 2937$ kJ/mol
- Under cellular conditions, ΔG for ATP hydrolysis ≈ 50 kJ/mol
- 32 ATP /glucose = 1,600 kJ/mol
- Efficiency 1600/2937 X 100 = 54%
- 3.5 billion years of evolution has resulted in 54 % efficiency.

Photosynthesis

- The vast majority of energy consumed by living organisms stems from solar energy captured by the process of photosynthesis.
- Of the 1.5×10^{22} kJ of energy reaching the earth each day from the sun, 1% is absorbed by photosynthetic organisms and transduced into chemical energy.
- This energy, in the form of biomolecules, becomes available to other members of the biosphere through food chains.
  - The transduction of solar, or light, energy into chemical energy is often expressed in terms of **carbon dioxide fixation,** in which hexose is formed from carbon dioxide and oxygen is evolved



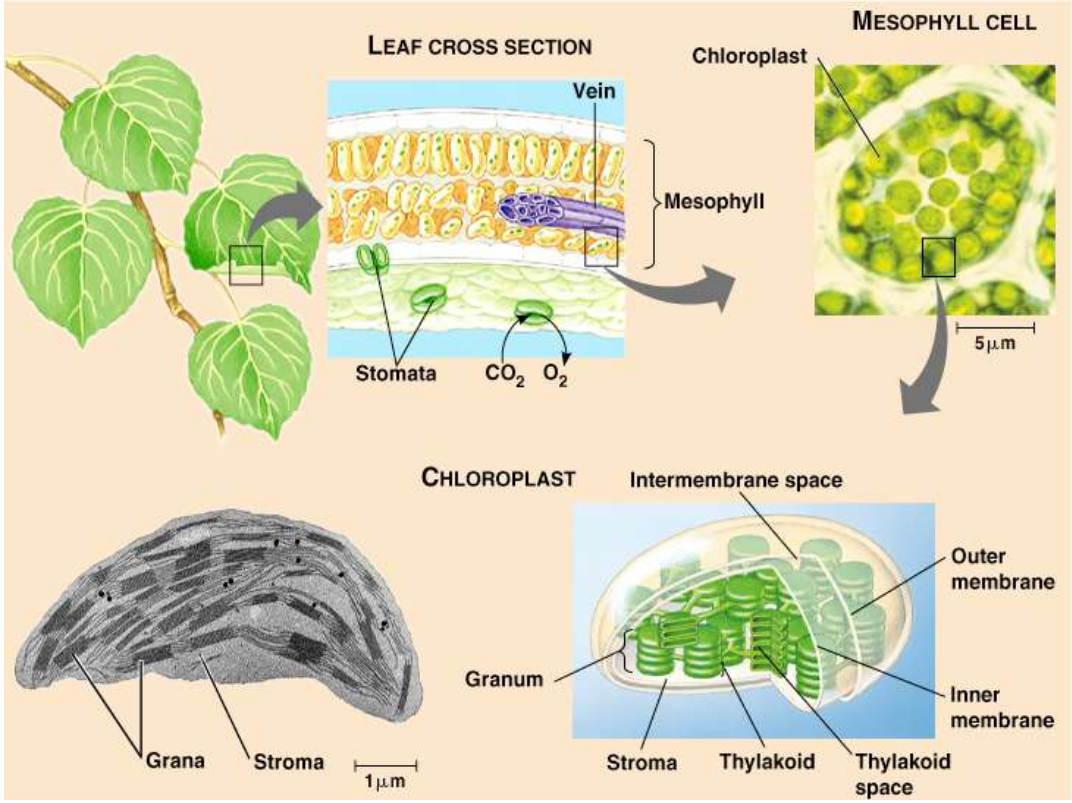


General Aspects of Photosynthesis

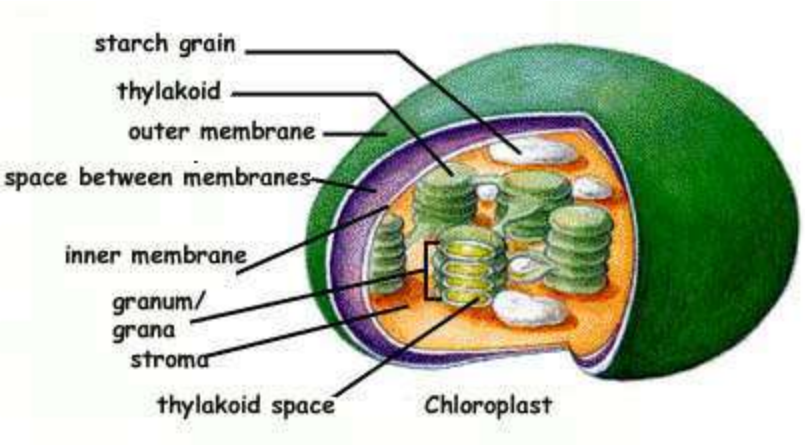
- Organisms capable of photosynthesis are very diverse, ranging from simple prokaryotic forms to the largest organisms of all, *Sequoia gigantea*, the giant redwood trees of California.
- An important general aspect is that *photosynthesis occurs in membranes*.
  - In photosynthetic prokaryotes, the photosynthetic membranes fill up the cell interior; in photosynthetic eukaryotes, the photosynthetic membranes are localized in large organelles known as **chloroplasts**
- Chloroplasts are one member in a family of related plant-specific organelles known as **plastids.**
  - Chloroplasts themselves show a range of diversity, from the single, spiral chloroplast that gives *Spirogyra* its name to the multitude of ellipsoidal plastids typical of higher plant cells

Electron micrograph of a representative chloroplast.

- All chloroplasts have the organization of the inner membrane system, **thylakoid membrane.**
- The thylakoid membrane is organized into paired folds that extend throughout the organelle
- These paired folds, or **lamellae**, give rise to flattened sacs or disks, **thylakoid vesicles** (from the Greek *thylakos*, meaning "sack"), which occur in stacks called **grana.**
- A single stack, or **granum,** may contain dozens of thylakoid vesicles, and different grana are joined by lamellae that run through the soluble portion, or **stroma,** of the organelle.

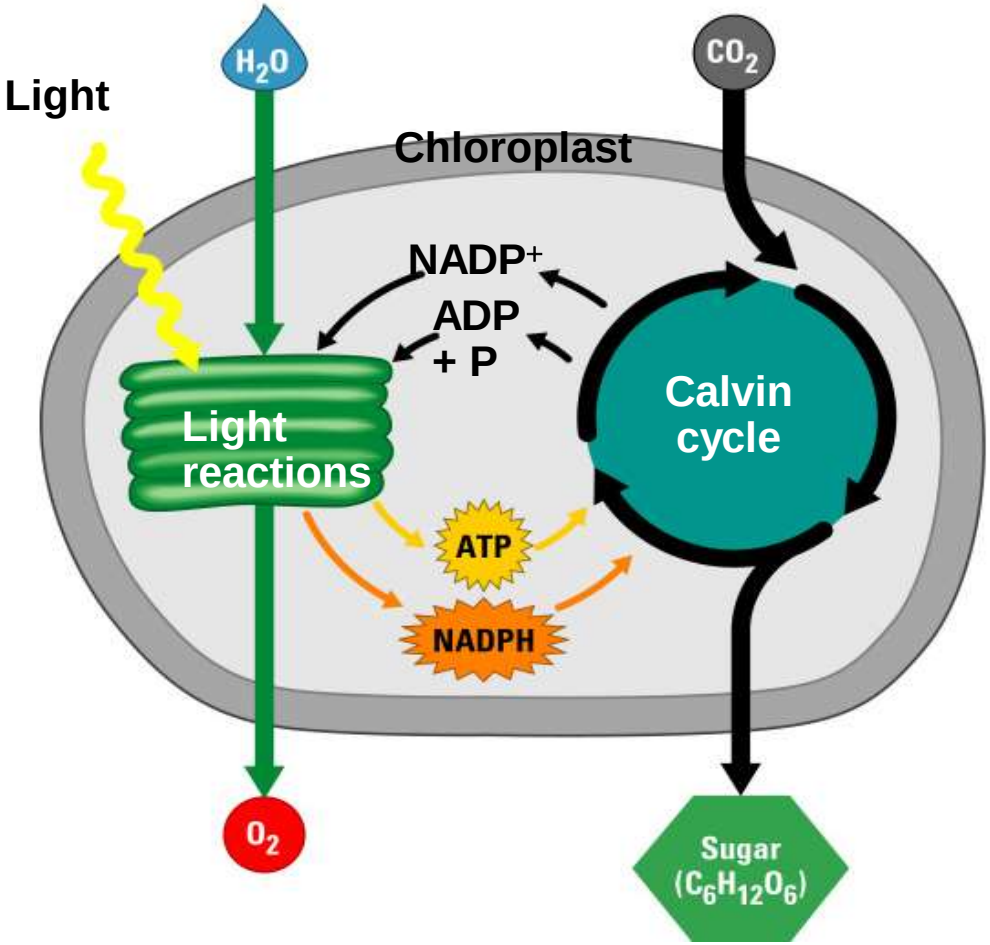


- Chloroplasts thus possess three membrane-bound aqueous compartments:
  - the intermembrane space,
  - the stroma, and
  - the interior of the thylakoid vesicles, the so-called **thylakoid space** (also known as the **thylakoid lumen**).
- This third compartment serves an important function in the transduction of light energy into ATP formation.
  - The thylakoid membrane has a highly characteristic lipid composition and, like the inner membrane of the mitochondrion, is impermeable to most ions and molecules.
  - Chloroplasts, like their mitochondrial counterparts, possess DNA, RNA, and ribosomes and consequently display a considerable amount of autonomy.
  - However, many critical chloroplast components are encoded by nuclear genes, so autonomy is far from absolute.



Photosynthesis Consists of Both Light Reactions and Dark Reactions

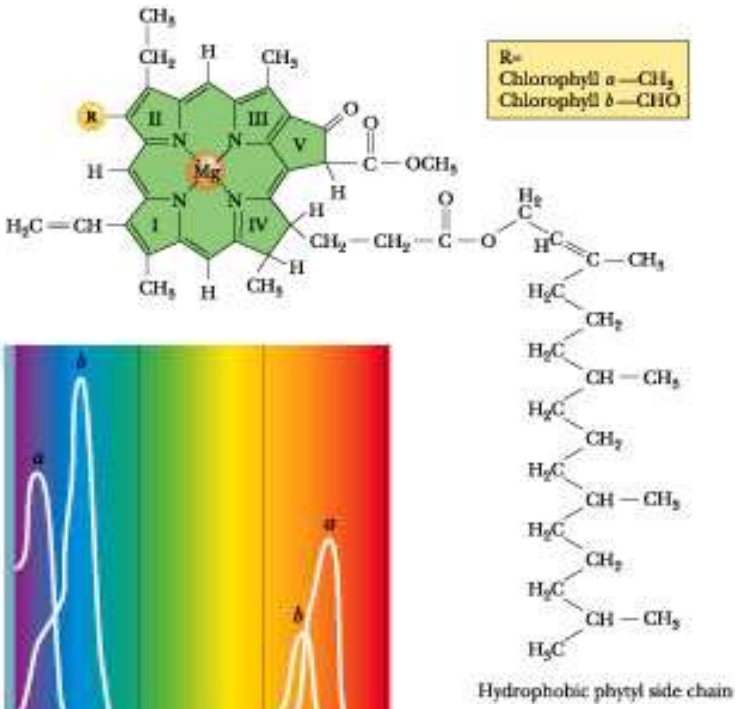
- If a chloroplast suspension is illuminated in the absence of carbon dioxide, oxygen is evolved.
- If the illuminated chloroplasts are now placed in the dark and supplied with CO_2, net hexose synthesis can be observed.
- The **light reactions** of photosynthesis, of which O_2 evolution is only one part, are associated with the thylakoid membranes. In contrast, the light-independent reactions or **dark reactions,** notably CO_2 fixation, are located in the stroma.

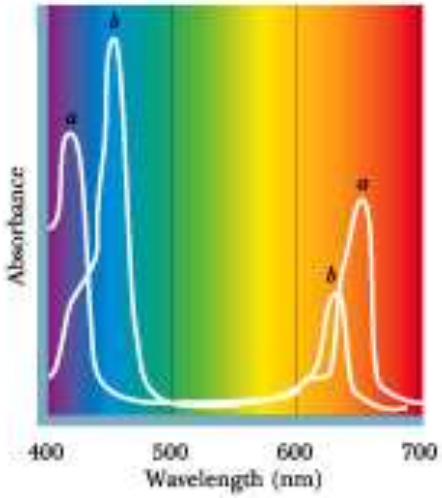


- Light is transformed by thylakoids to yield chemical energy in the form of reducing potential (NADPH) and high-energy phosphate (ATP).
- NADPH and ATP can then be used to drive the endergonic process of hexose formation from CO_2 in the stroma

Photosynthesis Depends on the Photoreactivity of Chlorophyll

- **Chlorophylls (*a* and *b*)** are Mg-containing substituted tetrapyrroles, similar to heme, the Fe-containing porphyrin
  - Chlorophylls differ from heme in a number of properties: magnesium instead of iron is coordinated in the center of the planar conjugated ring structure; a long-chain alcohol, **phytol,** is esterified to a pyrrole ring substituent; and the methine bridge linking pyrroles III and IV is substituted and cross-linked to ring III, leading to the formation of a fifth five-membered ring.

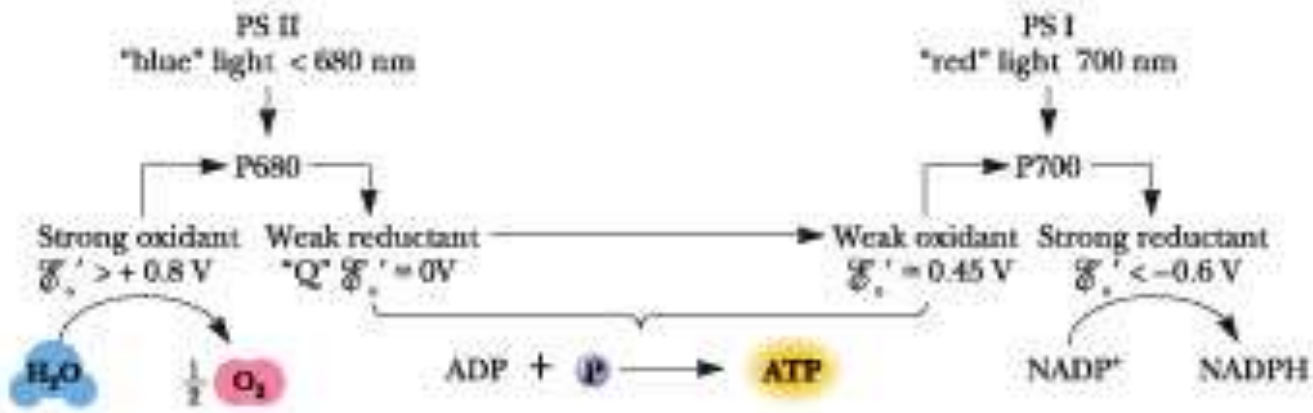




- Chlorophylls are excellent light absorbers because of their aromaticity.
- Other pigments in photosynthetic organisms, so-called **accessory light-harvesting pigments**
  - *carotenoids* and *phycobilins*, are also responsible for the magnificent colors of autumn.

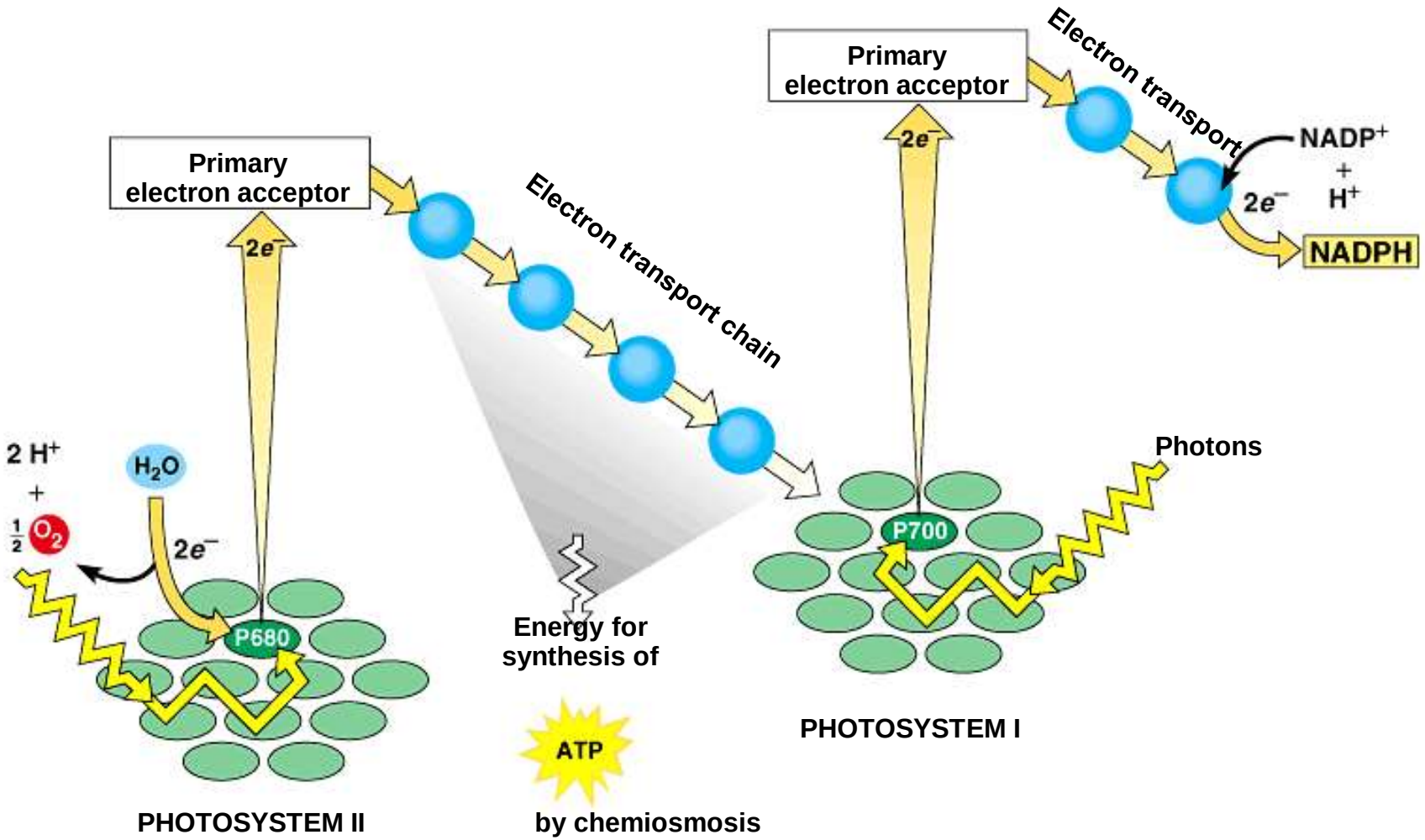
Eukaryotic Phototrophs Possess Two Distinct Photosystems

- Two separate but interacting photosystems exist in photosynthetic eukaryotes
- The existence of two light reactions established the presence of two photosystems, I and II.
- **Photosystem I** (PSI) is defined as containing reaction center chlorophylls with maximal red light absorption at 700 nm; PSI is not involved in O_2 evolution.
- **Photosystem II** (PSII) functions in O_2 evolution, using reaction centers that exhibit maximal red light absorption at 680 nm.



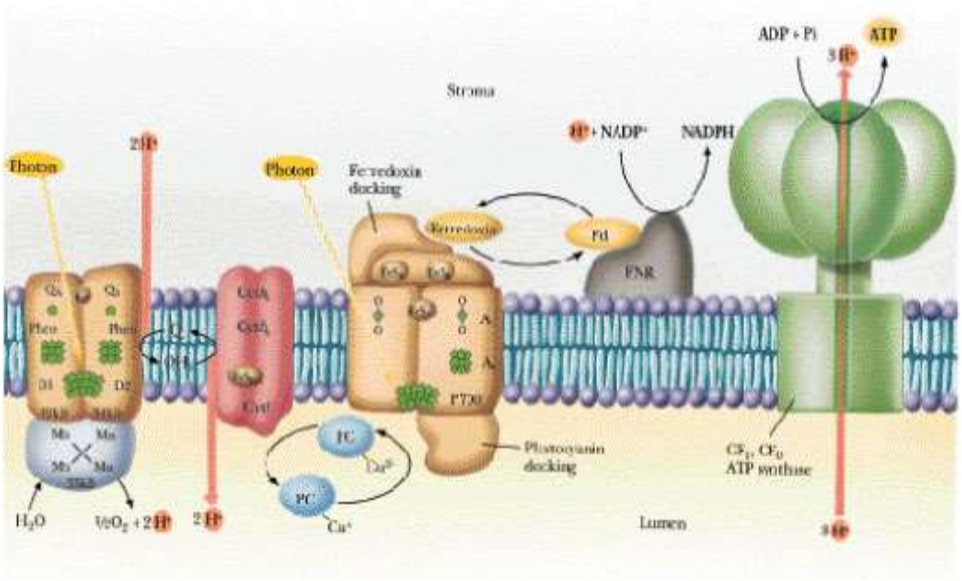
Roles of the two photosystems, PSI and PSII.

- Photosystem I provides reducing power in the form of NADPH.
- Photosystem II splits water, producing O_2, and feeds the electrons released into an electron transport chain that couples PSII to PSI.
- Electron transfer between PSII and PSI pumps protons for chemiosmotic ATP synthesis.



Photophosphorylation- Light-Driven ATP Synthesis

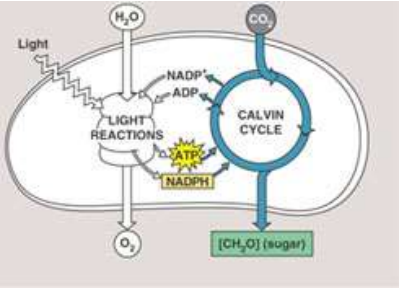
- Light-driven ATP synthesis, termed photophosphorylation, is a fundamental part of the photosynthetic process.
- The conversion of light energy to chemical energy results in electron-transfer reactions leading to the generation of reducing power (NADPH).
- Coupled with these electron transfers, protons are driven across the thylakoid membranes from the stromal side to the lumenal side.
- These proton translocations occur in a manner similar to the proton translocations accompanying mitochondrial ETC

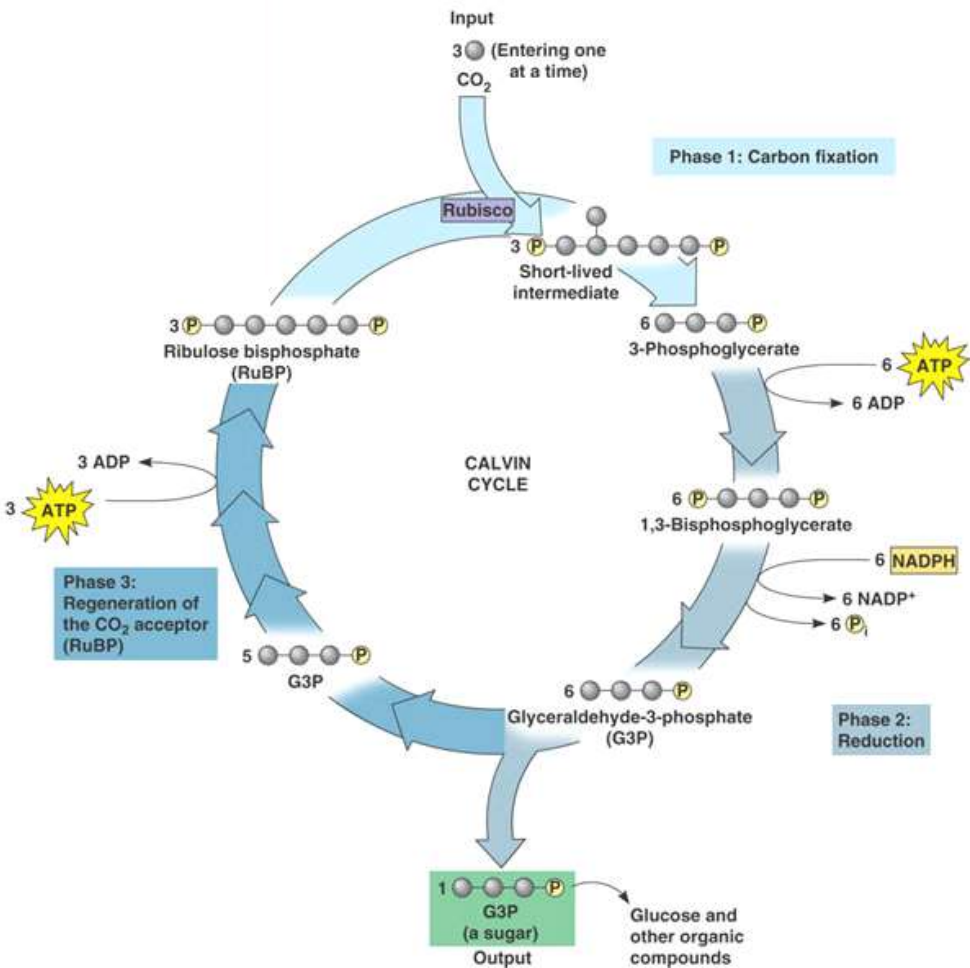


Photosynthetic electron transport establishes a proton gradient that is tapped by the CF_1CF_0 ATP synthase to drive ATP synthesis. Critical to this mechanism is the fact that the membrane-bound components of light-induced electron transport and ATP synthesis are asymmetrical with respect to the thylakoid membrane so that vectorial discharge and uptake of H^1 ensue, generating the proton-motive force.

Carbon Dioxide Fixation: Calvin-Benson Cycle

- **Calvin Cycle**, earlier designated the photosynthetic "dark reactions," is now called the **carbon reactions** pathway:
  - The free energy of cleavage of ~P bonds of **ATP**, and reducing power of **NADPH**, are used to fix and reduce **CO_2** to form **carbohydrate**.
  - Enzymes & intermediates of the **Calvin Cycle** are located in the chloroplast **stroma**, a compartment somewhat analogous to the mitochondrial matrix.



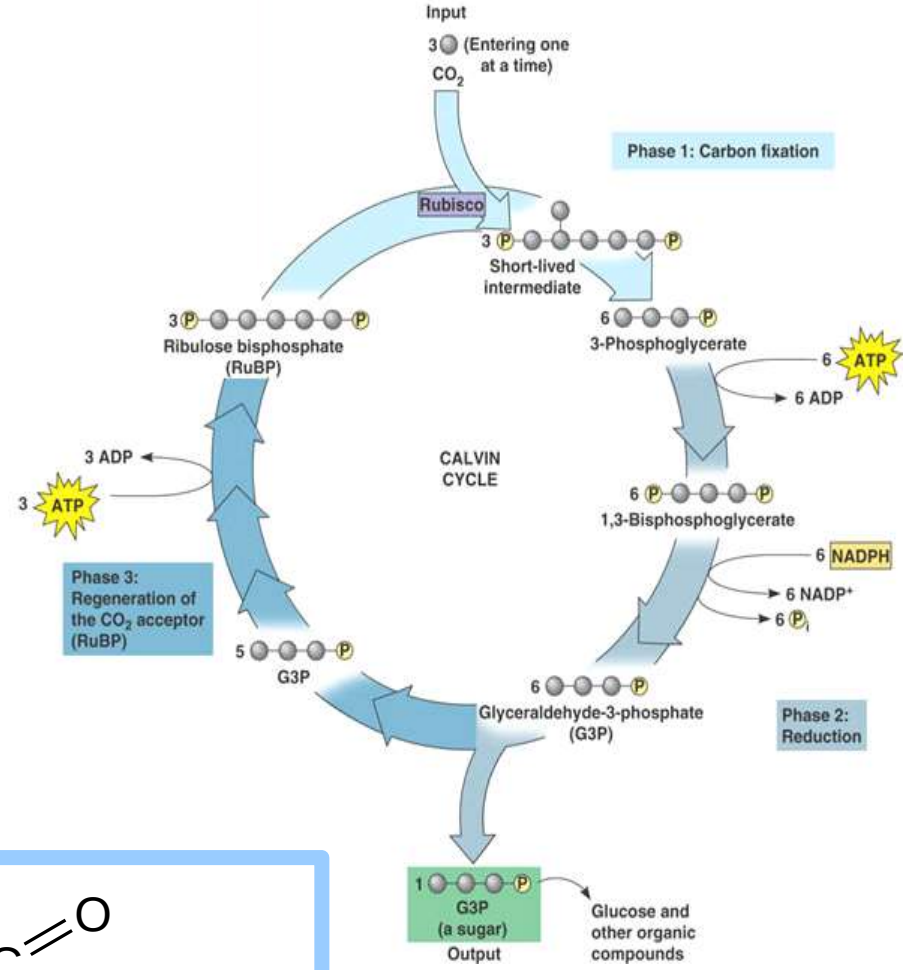


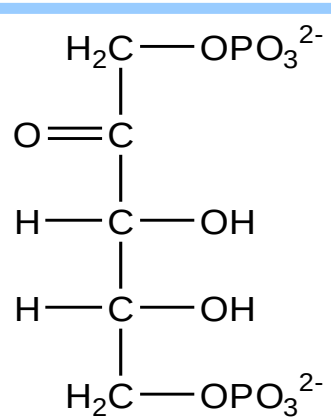
STEP 1:

- **Ribulose Bisphosphate Carboxylase** (RuBP Carboxylase), catalyzes CO_2 fixation:

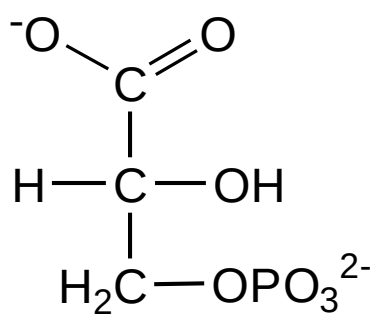
ribulose-1,5-bisphosphate + CO_2 → 2 3-phosphoglycerate

- Because it can alternatively catalyze an oxygenase reaction, the enzyme is also called RuBP Carboxylase/Oxygenase (**RuBisCO**). It is the **most abundant** enzyme on earth.





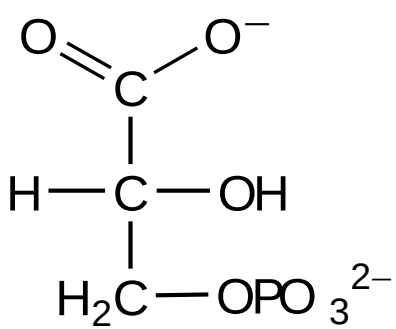
Ribulose-1,5-bisphosphate (RuBP)



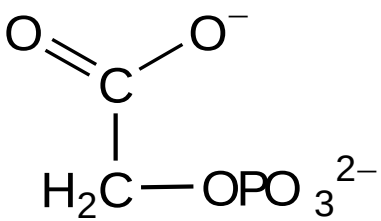
3-Phosphoglycerate (3PG)

Photorespiration:

- **O_2** can **compete with CO_2** for binding to RuBisCO, especially when [CO_2] is low & [O_2] is high.
- When **O_2** reacts with ribulose-1,5-bisphosphate, the products are 3-phosphoglycerate plus the **2-C** compound **2-phosphoglycolate**.
- This reaction is the basis for the name RuBP Carboxylase/Oxygenase (RuBisCO).
- **Photorespiration** is a **wasteful** process, substantially reducing efficiency of CO_2 fixation, even at normal ambient CO_2.

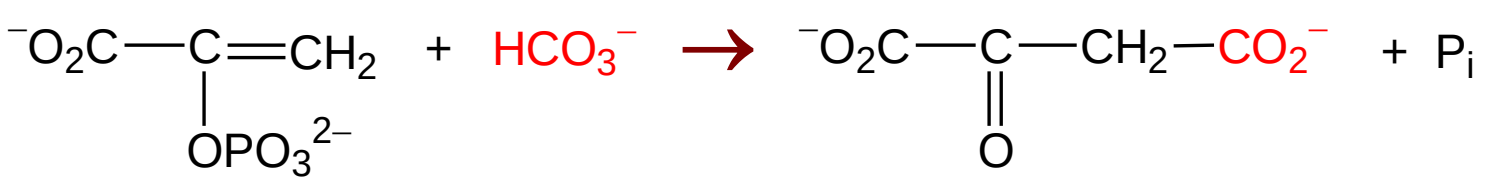


3-phospho-glycerate



phosphoglycolate

- Most plants, designated **C_3**, fix CO_2 initially via RuBP Carboxylase, yielding the 3-C 3-phosphoglycerate.
- Plants designated **C_4** have one cell type in which phosphoenolpyruvate (PEP) is carboxylated via the enzyme PEP Carboxylase, to yield the **4-C oxaloacetate**.
  - Oxaloacetate is converted to other 4-C intermediates that are transported to cells active in photosynthesis, where **CO_2** is **released** by decarboxylation.
- **C_4 plants maintain a high ratio of CO_2/O_2** within photosynthetic cells, ***thus minimizing photorespiration***.
- Research has been aimed at increasing expression of and/or inserting genes for C_4 pathway enzymes, such as PEP Carboxylase, in C_3 plants.



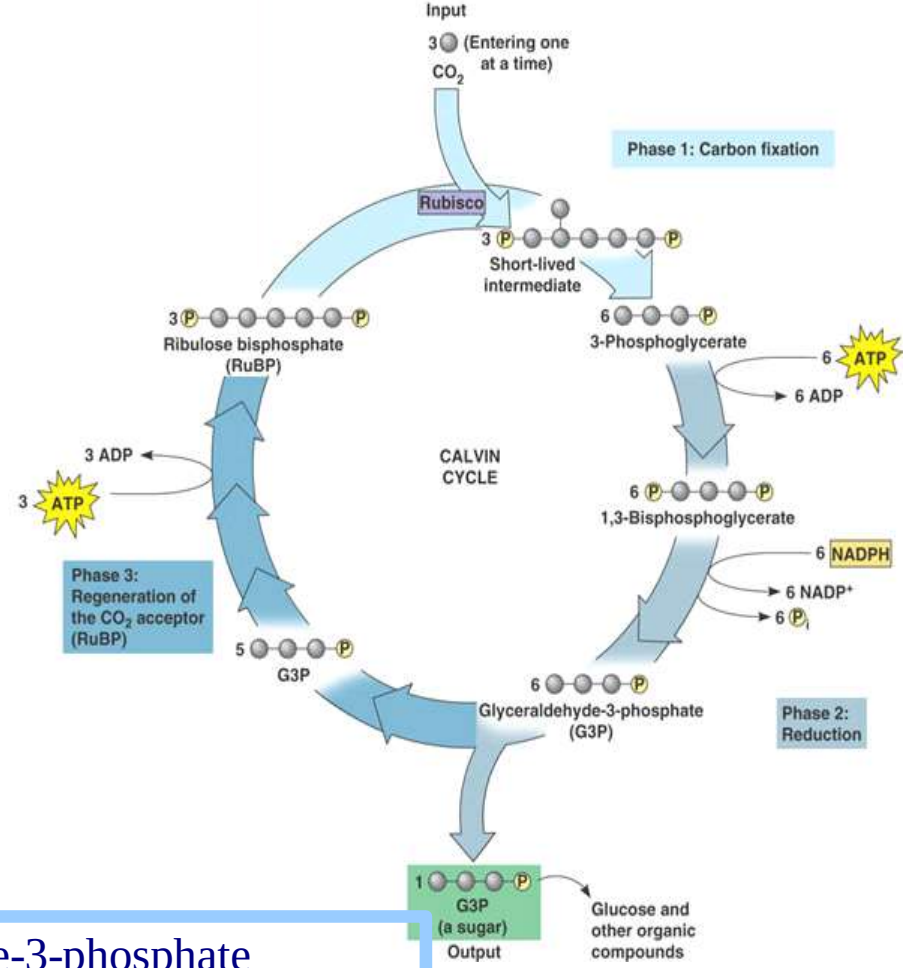
phosphoenolpyruvate (PEP)

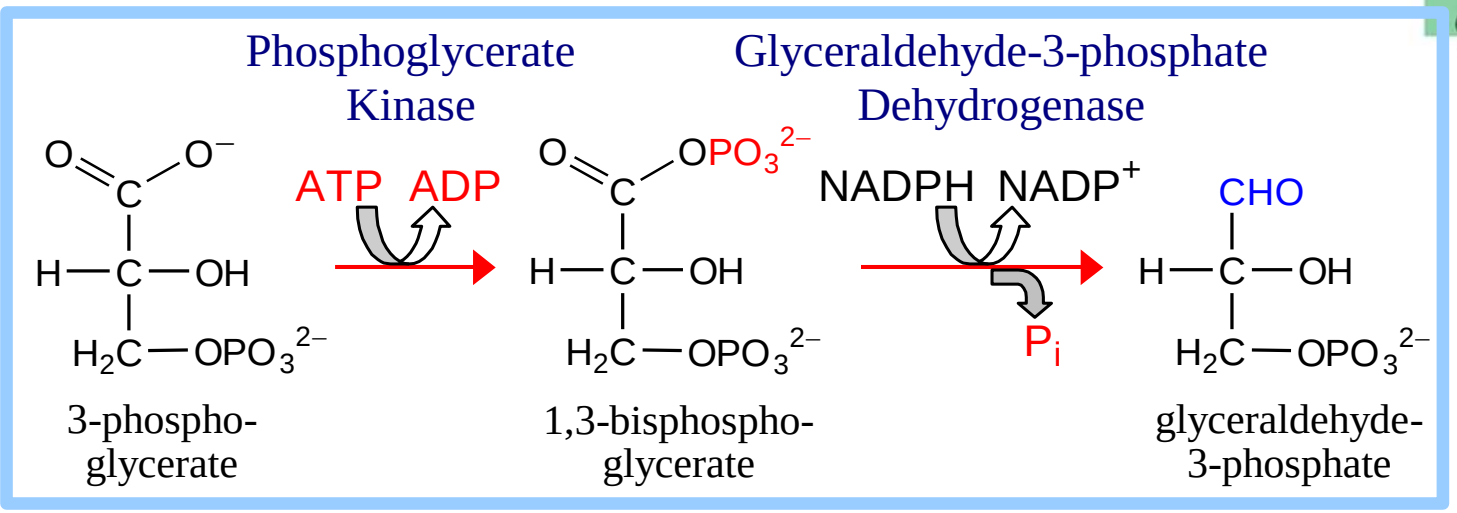
oxaloacetate

PEP Carboxylase

STEP 2:

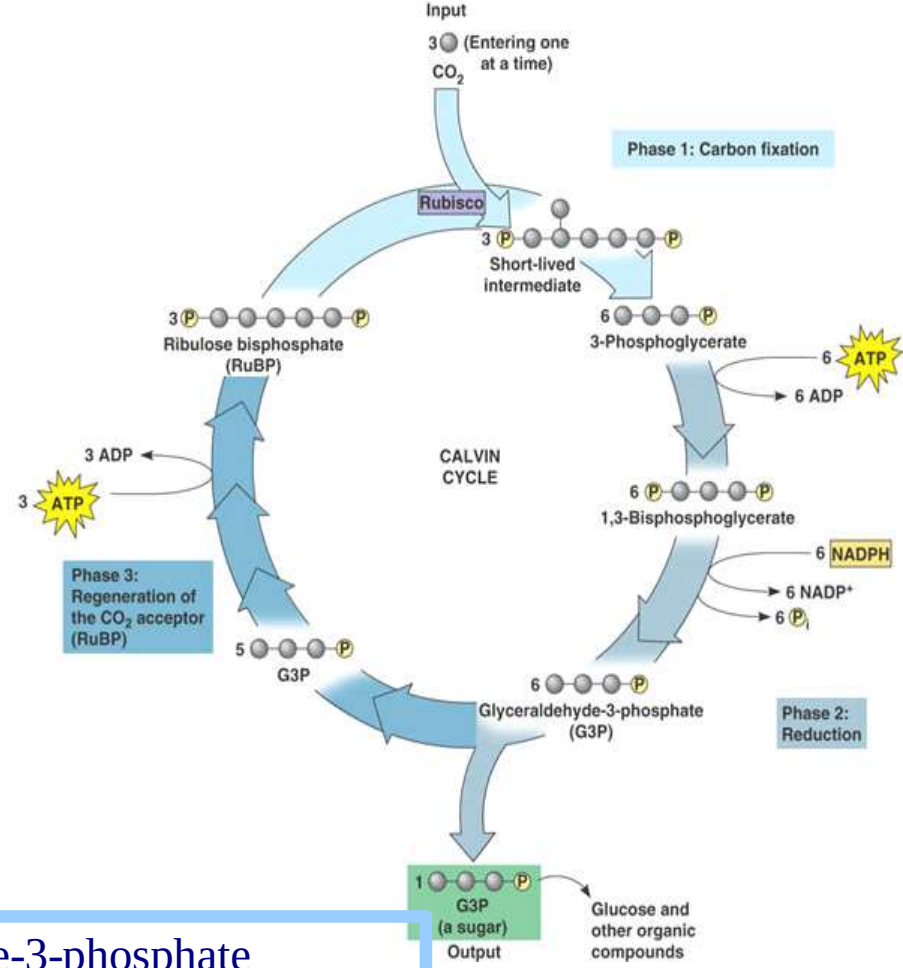
- The normal RuBP Carboxylase product, 3-phospho-glycerate is converted to glyceraldehyde-3-P.
- **Phosphoglycerate Kinase** catalyzes transfer of **P_i** from **ATP** to the carboxyl of **3-phosphoglycerate** (RuBP Carboxylase product) to yield **1,3-bisphosphoglycerate**.

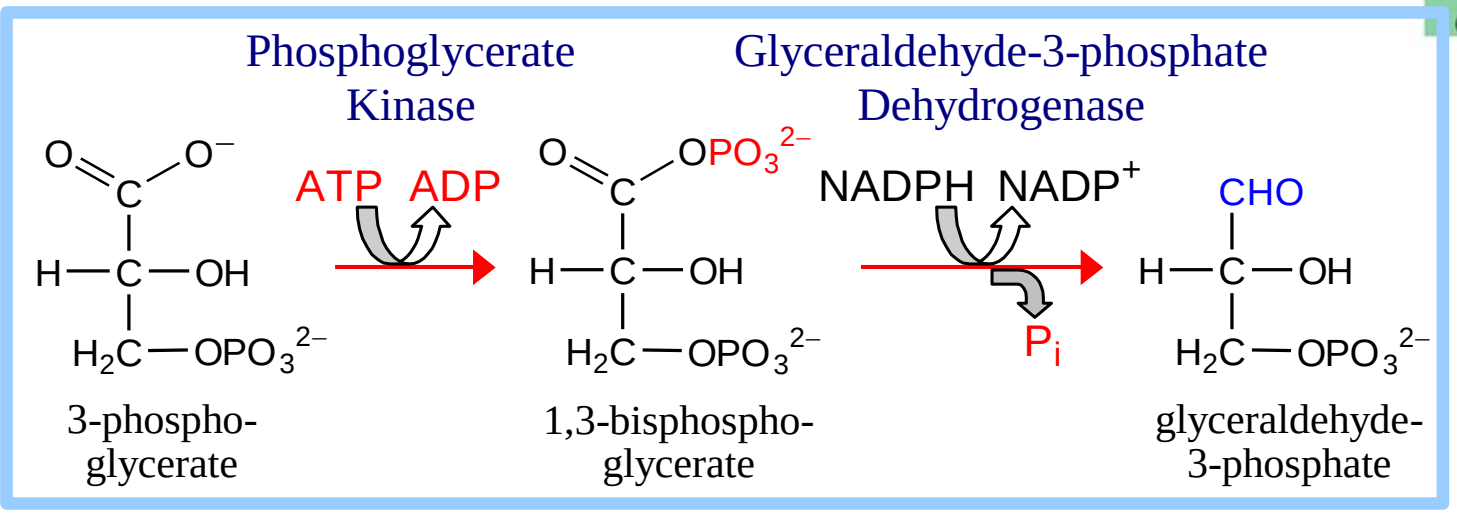




STEP 3:

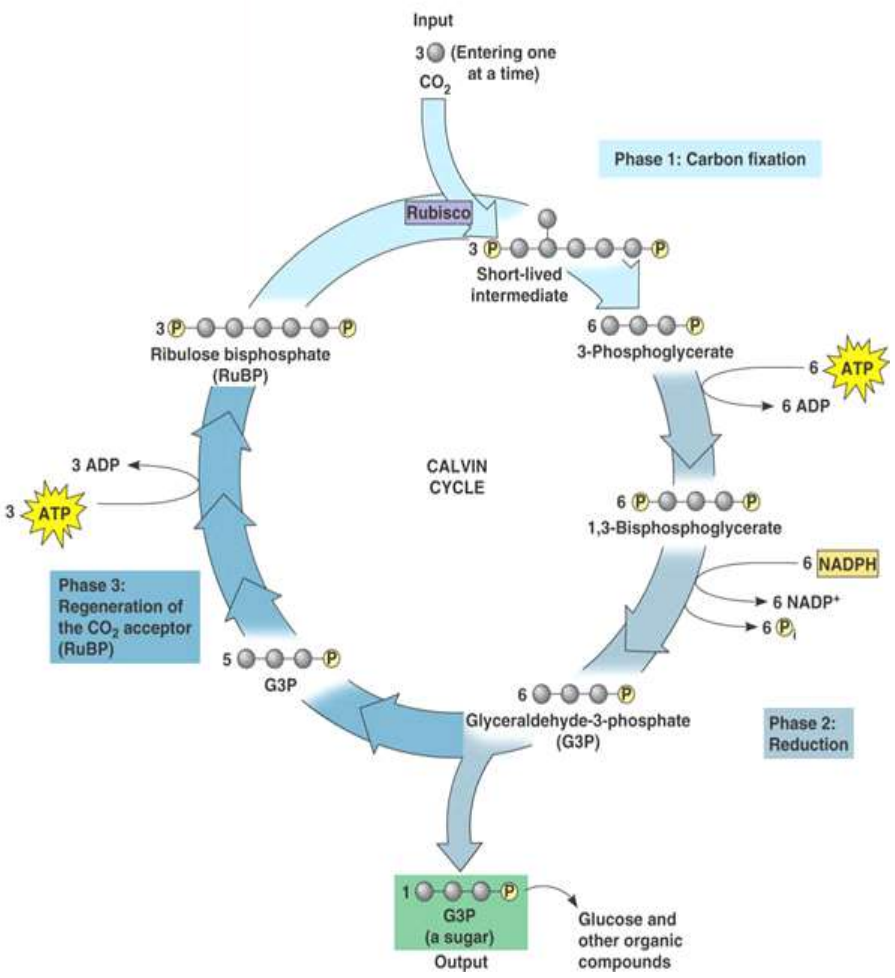
- **Glyceraldehyde-3-P Dehydrogenase** catalyzes reduction of the carboxyl of 1,3-bisphosphoglycerate to an **aldehyde**, with release of **P_i**, yielding **glyceraldehyde-3-P**.
- This is like the Glycolysis enzyme running backward, but the chloroplast Glyceraldehyde-3-P Dehydrogenase uses **NADPH** as e^- donor, while the cytosolic Glycolysis enzyme uses NAD^+ as e^- acceptor.





STEP 4:

- A portion of the **glyceraldehyde-3-P** is converted **back to ribulose-1,5-bisP**, the substrate for RuBisCO, via reactions catalyzed by:
  - Triose Phosphate Isomerase, Aldolase, Fructose Bisphosphatase, Sedoheptulose Bisphosphatase, Transketolase, Epimerase, Ribose Phosphate Isomerase, & Phosphoribulokinase.
- Many of these are similar to enzymes of Glycolysis, Gluconeogenesis or Pentose Phosphate Pathway, but are separate gene products found in the chloroplast stroma. (Enzymes of the other pathways listed are in the cytosol.)
- The process is similar to Pentose Phosphate Pathway run backwards.



Summary of Calvin cvcle

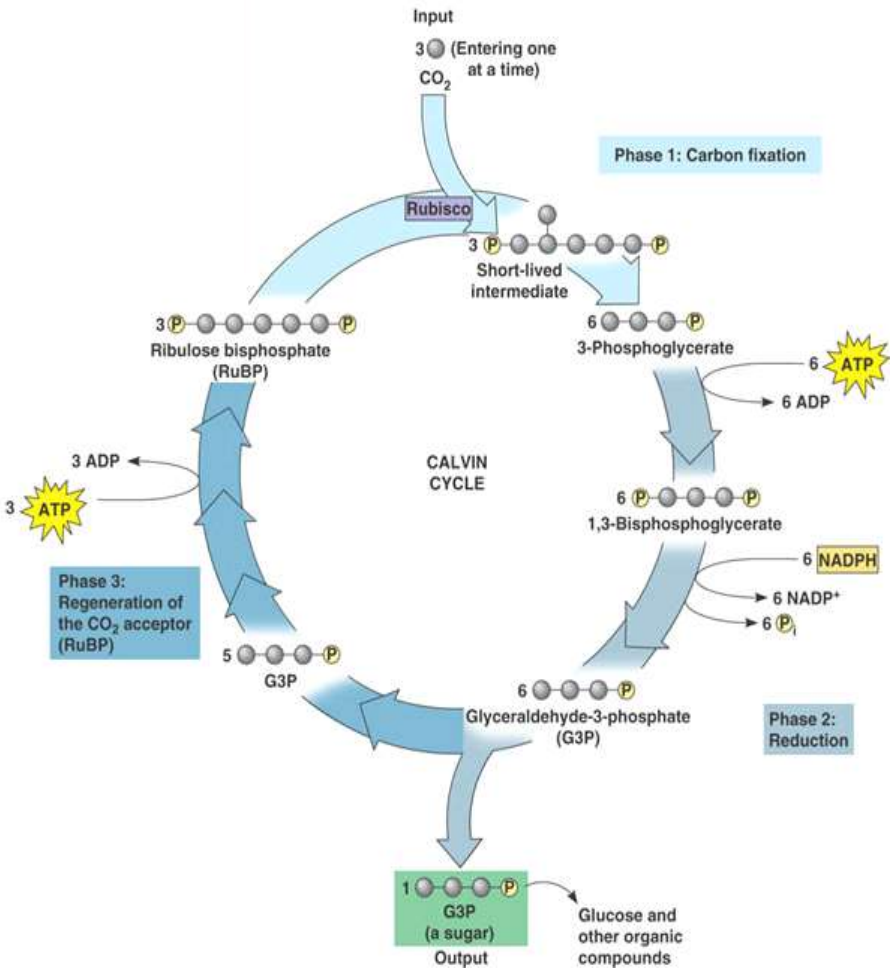
3 5-C ribulose-1,5-bisP (total of **15 C**) are carboxylated (**3 C** added), cleaved, phosphorylated, reduced, & dephosphorylated, yielding

6 3-C glyceraldehyde-3-P (total of **18 C**). Of these:

- **1 3-C** glyceraldehyde-3-P exits as **product**.
- **5 3-C** glyceraldehyde-3-P (**15 C**) are recycled back into **3 5-C** ribulose-1,5-bisphosphate.

$$C_3 + C_3 \rightarrow C_6$$
$$C_3 + C_6 \rightarrow C_4 + C_5$$
$$C_3 + C_4 \rightarrow C_7$$
$$C_3 + C_7 \rightarrow C_5 + C_5$$

Overall $5\ C_3 \rightarrow 3\ C_5$

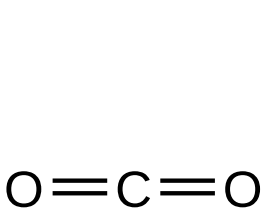


Summary of Calvin Cycle

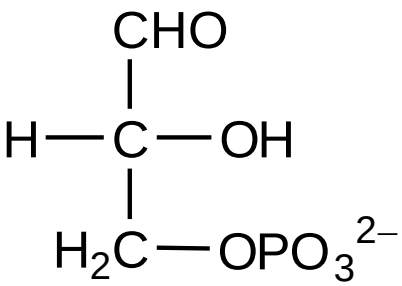
- **$3\ CO_2 + 9\ ATP + 6\ NADPH \rightarrow$ glyceraldehyde-3-P + 9 ADP + 8 P_i + 6 $NADP^+$**

- **Glyceraldehyde-3-P** may be converted to **other CHO**:
  - metabolites (e.g., fructose-6-P, glucose-1-P)
  - energy stores (e.g., sucrose, starch)
  - cell wall constituents (e.g., cellulose).

- **Glyceraldehyde-3-P** can also be utilized by plant cells as carbon source for synthesis of **other compounds** such as fatty acids & amino acids.



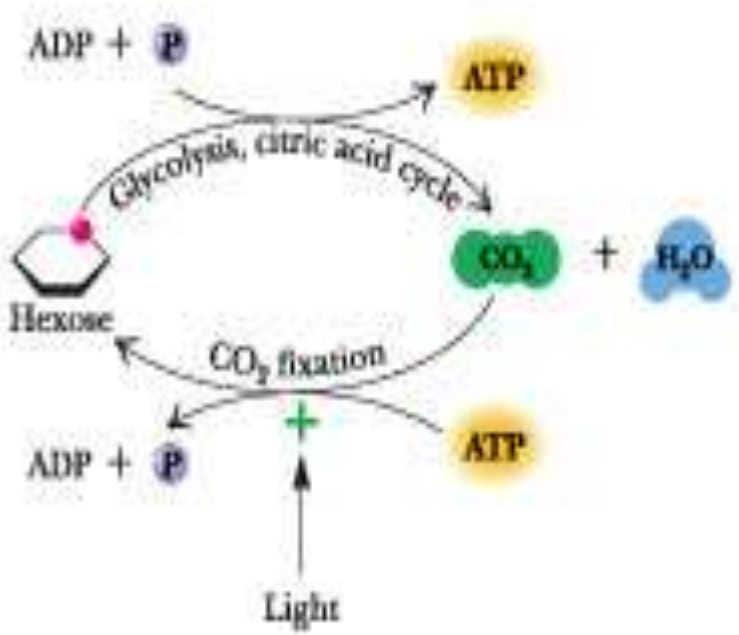
carbon dioxide

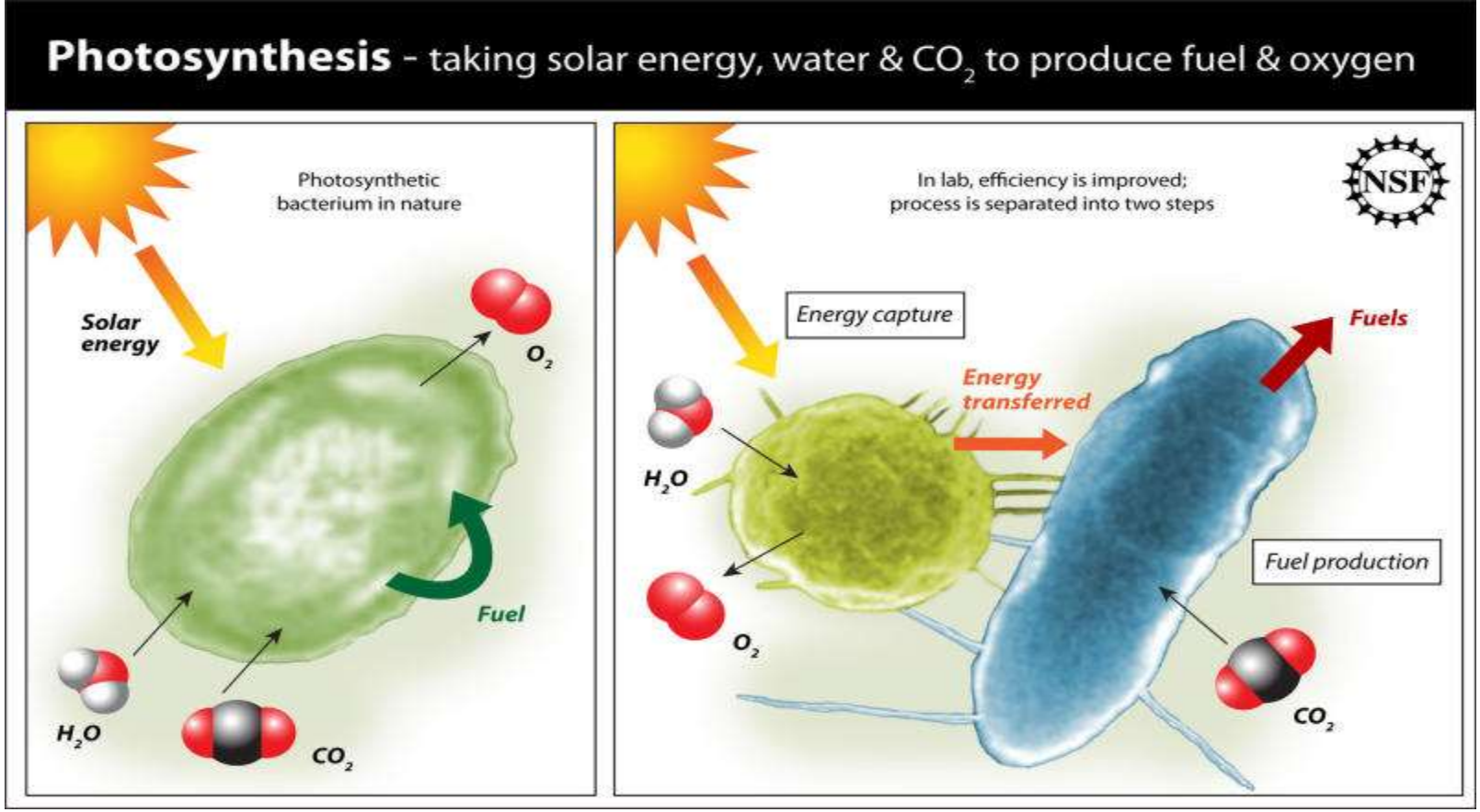


glyceraldehyde-3-phosphate

Regulation of Carbon Dioxide Fixation

- When **light energy** is available to generate ATP and NADPH for CO_2 fixation, the **Calvin cycle proceeds**.
- The light-induced changes in the chloroplast which regulate key Calvin cycle enzymes include
  - (1) *changes in stromal pH,*
  - (2) *generation of reducing power, and*
  - (3) *Mg^{2+} efflux from the thylakoid lumen.*
- In the **dark**, when ATP and NADPH cannot be produced by photosynthesis, **fixation of CO_2 ceases**.
  - Plant cells contain mitochondria and can carry out cellular respiration (glycolysis, the citric acid cycle, and oxidative phosphorylation) to provide energy in the dark through futile cycling of carbohydrate to CO_2





•*In photosynthesis, solar energy is captured and used to produce chemical fuel by a photosynthetic organism. This project is designed to improve the efficiency of this capture and conversion by: 1) separating them into two types of cells: one that captures solar energy and another "factory" cell that produces fuel; and 2) enabling these two different types of cells to communicate with one another via the flow of electrical currents between them. Compartmentalizing the processes of energy capture and fuel production into two different types of cells will allow researchers to optimize environments for each process, and thereby improve the efficiency of each process. (Credit: Zina Deretsky, NSF)*